

EDEXCEL GCSE BUSINESS

CALCULATION PRACTICE WORKBOOK



STUDENT NAME:

Section 1: Formula for Calculations

The list below provides you with the names of the key calculations which you may need to use during the examinations for Paper 1 or Paper 2 of the Edexcel GCSE Business qualification.

Write the correct formula for each of the key calculations in the boxed provided below.

Total costs (1)

Revenue (1)

Break-even point in units (1)

Break-even point in costs / revenue (1)

Margin of safety (1)

Interest (on loans) (1)

Net cash-flow (1)

Opening balance (1)

Closing balance (1)

Gross profit (1)

Gross profit margin (1)

Net profit (1)

Net profit margin (1)

Average rate of return (1)

Section 1 score: /14

Key Things to Remember About Formula for Calculations

Write down any key information that you need to remember about formulas for calculations in this box.

Section 2: Percentages and Percentage Changes

1. A company's revenue was £500,000 last year and it increased to £600,000 this year. Calculate the percentage increase in revenue. You are advised to show your workings (2)

2. A store purchased a product for £50 and sold it the following month for 50% more than this cost price. Calculate the amount the store sold the product for. You are advised to show your workings (2)

3. A store's revenue in January was £50,000, in February, it increased to £60,000, and in March, it further increased to £75,000. Calculate the percentage increase in revenue from February to March. You are advised to show your workings (2)

4. In a survey of 800 customers, 500 stated that they prefer Product A, 200 prefer Product B, and the remaining customers have no preference. What is the percentage of customers who prefer Product A? You are advised to show your workings (2)

5. Out of a total of 1,500 job applicants, 900 were qualified for the position, 300 were underqualified, and the rest were overqualified. What percentage of applicants were qualified? You are advised to show your workings (2)

6. An employee received a salary raise from £40,000 per year to £45,000 per year. Calculate the percentage increase in the salary. You are advised to show your workings (2)

7. A company had 200 employees but planned to layoff 20% of its workforce. Calculate the number of employees that would remain after the layoffs. You are advised to show your workings (2)

8. A store had 500 units of a product in stock and it sold 150 units. Calculate the percentage of units sold. You are advised to show your workings (2)

9. A company sold 8,000 units of Product X in a month. If the total market for that product was estimated to be 40,000 units, and their main competitor sold 12,000 units, what is the company's market share as a percentage? You are advised to show your workings (2)

- 10.** Out of a total revenue of £1,000,000, £300,000 was generated from online sales, £500,000 from in-store sales, and the rest from wholesale. What percentage of the total revenue came from online sales? You are advised to show your workings (2)

Section 2 score: /20

Key Things to Remember About Percentages and Percentage Changes

Write down any key information that you need to remember about percentages and percentage changes in this box.

Section 3: Averages

1. A bakery sells 100 loaves of bread in a week. The prices of the loaves sold are as follows: £2.50, £3.00, £2.75, £3.25, and £2.90. Calculate the average price of the loaves. You are advised to show your workings (2)

2. A store offers a summer sale on t-shirts. The prices of the t-shirts are as follows: £12.99, £9.99, £14.99, £11.99, and £13.99. Calculate the average price of the t-shirts. You are advised to show your workings (2)

3. A car rental company rents out four cars in a day. The rental prices of the cars are as follows: £45.00, £50.00, £55.00, and £48.00. Calculate the average rental price of the cars. You are advised to show your workings (2)

4. A café sells three different types of coffee: regular, decaf, and espresso. The number of cups sold in a day is as follows: regular (60 cups), decaf (40 cups), and espresso (30 cups). Calculate the average number of cups sold per type of coffee. You are advised to show your workings (2)

5. A toy store sells 200 toys in a month. The sales revenue from the toys sold is as follows: £400, £500, £350, £450, and £600. Calculate the average sales revenue per toy. You are advised to show your workings (2)

6. A clothing store has three employees. Their monthly salaries are as follows: £1,800, £2,200, and £2,500. Calculate the average monthly salary of the employees.

7. A supermarket sells a variety of apples. The prices of the apples are as follows: £0.80, £0.75, £0.90, £0.70, and £0.85. Calculate the average price of the apples. You are advised to show your workings (2)

8. A logistics company has three trucks. The maintenance costs for each truck are as follows: £1,200, £1,500, and £1,800. Calculate the average maintenance cost per truck. You are advised to show your workings (2)

9. A delivery service has five drivers. The number of deliveries made by each driver in a week is as follows: 20 deliveries, 15 deliveries, 18 deliveries, 22 deliveries, and 25 deliveries. Calculate the average number of deliveries made per driver. You are advised to show your workings (2)

10. A restaurant offers three types of pizzas: Margherita, Pepperoni, and Vegetarian. The number of pizzas sold in a day is as follows: Margherita (40 pizzas), Pepperoni (30 pizzas), and Vegetarian (20 pizzas). Calculate the average number of pizzas sold per type. You are advised to show your workings (2)

Section 3 score: /20

Key Things to Remember About Averages

Write down any key information that you need to remember about averages in this box.

Section 4: Revenue

1. A bookstore sells 50 books at a price of £12 each. Calculate the sales revenue. You are advised to show your workings (2)

2. A clothing store sells 80 t-shirts at a price of £20 each. Calculate the sales revenue. You are advised to show your workings (2)

3. A restaurant serves 120 meals during lunchtime and generates £1,800 in sales revenue. What is the selling price per unit? You are advised to show your workings (2)

4. A tech store sells 30 smartphones at a price of £500 each. Calculate the sales revenue. You are advised to show your workings (2)

5. A supermarket sells 200 cans of pop and generates £300 in sales revenue. What is the selling price per unit? You are advised to show your workings (2)

6. A bakery generates £1,000 in sales revenue selling cakes for £25. Calculate how many cakes the bakery sold. You are advised to show your workings (2)

7. A car dealership sells 10 cars at an average price of £30,000 each. Calculate the sales revenue. You are advised to show your workings (2)

8. An online retailer sells 130 laptops and generates £44,200 in sales revenue. Calculate the selling price per unit. You are advised to show your workings (2)

9. A toy store sells 50 board games at an average price of £30 each. Calculate the sales revenue. You are advised to show your workings (2)

10. A coffee shop sells cups of coffee at a price of £3.50 each and generates a total of £525 in sales revenue. Calculate how many cups of coffee the coffee shop sold. You are advised to show your workings (2)

Section 4 score: /20

Key Things to Remember About Revenue

rite down any key information that you need to remember about revenue in this box.

Section 5: Variable Costs

1. A manufacturing company produces 500 units of a product. The variable cost per unit is £8. Calculate the total variable cost. You are advised to show your workings (2)

2. A clothing store purchases 100 t-shirts from a supplier. The variable cost per t-shirt is £6. Calculate the total variable cost. You are advised to show your workings (2)

3. A company produces 500 units of a product. The direct material cost per unit is £5, and the direct labour cost per unit is £3. Calculate the total variable cost. You are advised to show your workings (2)

4. A bakery produces 200 loaves of bread and has a total variable cost of £250. Calculate the variable cost per unit. You are advised to show your workings (2)

5. A restaurant serves 120 meals during dinner service. The variable cost per meal is £5. Calculate the total variable cost. You are advised to show your workings (2)

6. A car manufacturer produces 150 cars. The direct material cost per car is £15,000, and the direct labour cost per car is £8,000. Calculate the total variable cost. You are advised to show your workings (2)

7. A tech store purchases 50 smartphones from a supplier and has a total variable cost of £15,000. Calculate the variable cost per unit. You are advised to show your workings (2)

8. A supermarket sells cans of pop. The variable cost per can is £0.50 and the total variable cost is £250. Calculate how many cans the supermarket sold. You are advised to show your workings (2)

9. A car dealership purchases 20 cars from a manufacturer. The variable cost per car is £15,000. Calculate the total variable cost. You are advised to show your workings (2)

10. A toy store orders 150 board games from a supplier and has a total variable cost of £1,200. Calculate the variable cost per unit. You are advised to show your workings (2)

Section 5 score: /20

Key Things to Remember About Variable Costs

Write down any key information that you need to remember about variable costs in this box.

Section 6: Total Costs

1. A manufacturing company produces 500 units of a product. The fixed cost is £2,000 and the variable cost per unit is £8. Calculate the total cost. You are advised to show your workings (2)

2. A clothing store produces 200 t-shirts. The fixed cost is £1,500 and the variable cost per t-shirt is £4. Calculate the total cost. You are advised to show your workings (2)

3. A bakery produces 300 loaves of bread. The fixed cost is £1,000 and the variable cost per loaf is £1.50. Calculate the total cost. You are advised to show your workings (2)

4. A restaurant serves 150 meals during lunchtime. The fixed cost is £3,000 and the total cost is £4,500. Calculate the variable cost per meal. You are advised to show your workings (2)

5. A tech store produces 100 smartphones. The fixed cost is £5,000 and the variable cost per smartphone is £150. Calculate the total cost. You are advised to show your workings (2)

6. A supermarket stocks 1,000 cans of pop. The fixed cost is £500 and the variable cost per can is £0.30. Calculate the total cost. You are advised to show your workings (2)

7. A car dealership sells 50 cars. The total cost is £135,000 and the variable cost per car is £2,500. Calculate the fixed cost.

8. A toy store produces 400 toys. The fixed cost is £2,500 and the variable cost per toy is £3. Calculate the total cost. You are advised to show your workings (2)

9. A coffee shop prepares 200 cups of coffee. The total cost is £1,100 which includes fixed costs of £1,000. Calculate the variable cost per cup. You are advised to show your workings (2)

10. A restaurant serves 80 pizzas. The fixed cost is £4,000 and the variable cost per pizza is £5. Calculate the total cost. You are advised to show your workings (2)

Section 6 score: /20

Key Things to Remember About Total Costs

Write down any key information that you need to remember about total costs in this box.

Section 7: Gross Profit

1. A bookstore sells 200 books at a price of £10 each. The unit cost of purchasing the books is £7.50. Calculate the gross profit. You are advised to show your workings (2)

2. A clothing store sells 100 t-shirts at a price of £15 each. The unit cost of purchasing the t-shirts is £8. Calculate the gross profit. You are advised to show your workings (2)

3. A restaurant serves 150 meals during lunchtime. The average price per meal is £10 and the business made a gross profit of £300. Calculate the cost of sales. You are advised to show your workings (2)

4. A tech store sells 50 smartphones at a price of £500 each. The unit cost of purchasing the smartphones is £400. Calculate the gross profit. You are advised to show your workings (2)

5. A supermarket sells 500 cans of pop and the unit cost of purchasing the cans is £0.60. The business makes £200 gross profit. Calculate the sales revenue. You are advised to show your workings (2)

6. A bakery sells 100 loaves of bread at a price of £3 each. The unit cost of producing each loaf of bread is £1.50. Calculate the gross profit. You are advised to show your workings (2)

7. A car dealership sells 10 cars at an average price of £25,000 each. The unit cost of purchasing each car is £18,000. Calculate the gross profit. You are advised to show your workings (2)

8. A toy store sells 200 toys at an average price of £20 each and makes a gross profit of £1,600. Calculate unit cost of purchasing each toy. You are advised to show your workings (2)

9. A coffee shop sells 300 cups of coffee at a price of £2.50 each. The unit cost of preparing each cup of coffee is £1. Calculate the gross profit. You are advised to show your workings (2)

10. A restaurant serves 80 pizzas. The average price per pizza is £12. The unit cost of preparing each pizza is £6. Calculate the gross profit. You are advised to show your workings (2)

Section 7 score: /20

Key Things to Remember About Gross Profit

Write down any key information that you need to remember about gross profit in this box.

Section 8: Gross Profit Margin

1. A bookstore sells 600 books at a price of £8 each. The unit cost of purchasing the books is £6. Calculate the gross profit margin. You are advised to show your workings (2)

2. A clothing store sells 2,000 t-shirts at a price of £12 each. The unit cost of purchasing the t-shirts is £9. Calculate the gross profit margin. You are advised to show your workings (2)

3. The table below contains information about a bakery. Calculate the gross profit margin for the bakery. You are advised to show your workings (2)

Business	Sales Revenue	Cost of Sales	Gross Profit
Bakery	£100,000	£60,000	£40,000

4. A restaurant serves 300 meals during lunchtime. The average price per meal is £12. The unit cost of preparing each meal is £5.50. Calculate the gross profit margin. You are advised to show your workings (2)

5. The table below contains information about a cosmetics company. Calculate the gross profit margin for the company. You are advised to show your workings (2)

Business	Sales Revenue	Cost of Sales	Gross Profit
Cosmetics	£420,000	£340,000	£80,000

6. A tech store sells 400 smartphones at a price of £300 each. The unit cost of purchasing the smartphones is £180. Calculate the gross profit margin. You are advised to show your workings (2)

7. A supermarket sells 500 cans of pop at a price of £1.20 each. The unit cost of purchasing the cans is £0.80. Calculate the gross profit margin. You are advised to show your workings (2)

8. A bakery sells 300 loaves of bread at a price of £3 each. The unit cost of producing each loaf of bread is £1.80. Calculate the gross profit margin. You are advised to show your workings (2)

9. A car dealership sells 60 cars at an average price of £24,000 each. The unit cost of purchasing each car is £21,000. Calculate the gross profit margin. You are advised to show your workings (2)

10. The table below contains information about a grocery store. Calculate the gross profit margin for the store. You are advised to show your workings (2)

Business	Sales Revenue	Cost of Sales	Gross Profit
Grocery	£900,000	£450,000	£450,000

Section 8 score: /20

Key Things to Remember About Gross Profit Margin

Write down any key information that you need to remember about gross profit margin in this box.

Section 9: Net Profit

1. A bookstore sells 2,500 books at a price of £15 each. The unit cost of purchasing the books is £9. Other operating expenses and interest amount to £5,500. Calculate the net profit. You are advised to show your workings (2)

2. A clothing store sells 160 t-shirts at a price of £22 each. The unit cost of purchasing the t-shirts is £14. Other operating expenses and interest amount to £450. Calculate the net profit. You are advised to show your workings (2)

3. A restaurant serves 80 meals during lunchtime. The average price per meal is £7. The unit cost of preparing each meal is £4.50. The business makes a net profit of £80. Calculate the other operating expenses and interest. You are advised to show your workings (2)

4. A tech store sells 30 smartphones at a price of £400 each. The unit cost of purchasing the smartphones is £300. Other operating expenses and interest amount to £1,000. Calculate the net profit. You are advised to show your workings (2)

5. A supermarket sells 700 cans of pop at a price of £2 each. The unit cost of purchasing the soda is £0.40. Other operating expenses and interest amount to £400. Calculate the net profit. You are advised to show your workings (2)

6. A bakery sells 50 loaves of bread at a price of £1 each. The unit cost of producing each loaf of bread is £0.70. Other operating expenses and interest amount to £300. Calculate the net profit. You are advised to show your workings (2)

7. A car dealership sells 15 cars at an average price of £20,000 each. The unit cost of purchasing each car is £12,000. Other operating expenses and interest amount to £85,000. Calculate the net profit. You are advised to show your workings (2)

8. A toy store sells 600 toys at an average price of £20 each. The unit cost of purchasing each toy is £14. Other operating expenses and interest amount to £1,500. Calculate the net profit. You are advised to show your workings (2)

9. A coffee shop sells 1,000 cups of coffee at a price of £2.50 each. The unit cost of preparing each cup of coffee is £0.75 Other operating expenses and interest amount to £1,800. Calculate the net profit. You are advised to show your workings (2)

10. A restaurant serves 350 pizzas. The average price per pizza is £8. The unit cost of preparing each pizza is £5. The business makes a net profit of £100. Calculate the other operating expenses and interest. You are advised to show your workings (2)

Section 9 score: /20

Key Things to Remember About Net Profit

Write down any key information that you need to remember about net profit in this box.

Section 10: Net Profit Margin

1. A bookstore sells 350 books at a price of £6 each. The unit cost of purchasing the books is £3.50. Other operating expenses and interest amount to £200. Calculate the net profit margin. You are advised to show your workings (2)

2. The table below contains information about a clothing Store. Calculate the net profit margin for the business. You are advised to show your workings (2)

Sales Revenue (£)	Gross Profit (£)	Other Operating Expenses (£)	Interest Expenses (£)
400,000	280,000	100,000	10,000

3. A clothing store sells 350 t-shirts at a price of £22 each. The unit cost of purchasing the t-shirts is £14. Other operating expenses and interest amount to £1,200. Calculate the net profit margin. You are advised to show your workings (2)

4. A restaurant serves 200 meals during lunchtime. The average price per meal is £3. The unit cost of preparing each meal is £1.20. Other operating expenses and interest amount to £250. Calculate the net profit margin. You are advised to show your workings (2)

5. The table below contains information about a business which offers consulting services. Calculate the net profit margin for the business. You are advised to show your workings (2)

Sales Revenue (£)	Gross Profit (£)	Other Operating Expenses (£)	Interest Expenses (£)
800,000	600,000	200,000	15,000

6. A tech store sells 60 smartphones at a price of £450 each. The unit cost of purchasing the smartphones is £300. Other operating expenses and interest amount to £4,600. Calculate the net profit margin. You are advised to show your workings (2)

7. A supermarket sells 5,000 cans of pop at a price of £2 each. The unit cost of purchasing the cans is £0.75. Other operating expenses and interest amount to £4,500. Calculate the net profit margin. You are advised to show your workings (2)

8. The table below contains information about a pharmacy. Calculate the net profit margin for the business. You are advised to show your workings (2)

Sales Revenue (£)	Gross Profit (£)	Other Operating Expenses (£)	Interest Expenses (£)
700,000	420,000	180,000	12,000

9. A bakery sells 600 loaves of bread at a price of £2.50 each. The unit cost of producing each loaf of bread is £1.30. Other operating expenses and interest amount to £650. Calculate the net profit margin. You are advised to show your workings (2)

10. A car dealership sells 22 cars at an average price of £46,000 each. The average unit cost of purchasing each car is £31,000. Other operating expenses and interest amount to £54,000. Calculate the net profit margin. You are advised to show your workings (2)

Section 10 score: /20

Key Things to Remember About Net Profit Margin

Write down any key information that you need to remember about net profit margin in this box.

Section 11: Sales, Costs, and Profit Tables

1. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)		6,500	7,200
Costs (£)	3,000		4,800
Profit (£)	2,000	2,000	

2. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)			
Costs (£)	4,500	5,500	6,000
Profit (£)	3,000	2,500	2,500

3. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)	4,000	4,500	4,800
Costs (£)			
Profit (£)	1,500	1,700	1,800

4. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)	6,200	7,000	7,800
Costs (£)	4,500	5,000	5,500
Profit (£)			

5. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)	3,500	4,000	
Costs (£)	2,000		2,700
Profit (£)		1,500	1,500

6. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)			
Costs (£)	5,000	5,500	6,000
Profit (£)	3,000	3,000	3,000

7. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)	4,500	5,000	5,200
Costs (£)			
Profit (£)	1,500	1,500	1,400

8. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)	6,800	7,200	7,500
Costs (£)	4,000	4,200	4,500
Profit (£)			

9. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)	5,200		6,000
Costs (£)		4,000	4,200
Profit (£)	1,400	1,500	

10. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1	Month 2	Month 3
Sales (£)	7,000		8,000
Costs (£)	4,500	4,800	
Profit (£)		2,700	3,000

Section 11 score: /30

Key Things to Remember About Sales, Costs, and Profit Tables

Write down any key information that you need to remember about sales, costs, and profit tables in this box.

Section 12: Average Rate of Return

1. A student invests £1,000 in a small online business and earns an average annual profit of £200. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

2. A student invests £10,000 in a business. Over a period of 7 years, the business generates total profits of £35,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

3. A restaurant invests £100,000 in expanding its operations. Over a period of 4 years, the expansion generates total profits of £80,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

4. A construction company invests £50,000 in a new project and earns an average annual profit of £7,500. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

5. A small business invests £25,000 in a marketing campaign. The campaign generates total profits of £40,000 over a period of 2 years. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

6. A manufacturing company invests £500,000 in a new production line. Over a period of 8 years, the production line generates total profits of £1,000,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

7. A restaurant invests £50,000 in a franchise and earns an average annual profit of £6,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

8. An entrepreneur invests £200,000 in a start-up venture. Over a period of 5 years, the venture generates total profits of £180,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

9. A retail store invests £20,000 in inventory and achieves an average annual profit of £4,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

10. A tech start-up invests £100,000 in research and development. Over a period of 4 years, the start-up generates total profits of £150,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

Section 12 score: /20

Key Things to Remember About Average Rate of Return

Write down any key information that you need to remember about average rate of return in this box.

Section 13: Net Cashflow

1. The table below contains cashflow information about a business. Calculate the **net cashflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£5,000	£8,000	£5,500		£7,500

2. The table below contains cashflow information about a business. Calculate the **cash outflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£10,000	£15,000		£3,000	£13,000

3. The table below contains cashflow information about a business. Calculate the **cash inflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£3,000		£2,000	£3,000	£6,000

4. The table below contains cashflow information about a business. Calculate the net cashflow for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£2,500	£6,000	£2,500		£6,000

5. The table below contains cashflow information about a business. Calculate the **cash outflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£15,000	£20,000		£5,000	£20,000

6. The table below contains cashflow information about a business. Calculate the **cash inflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£8,000		£7,000	£3,000	£11,000

7. The table below contains cashflow information about a business. Calculate the **net cashflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£12,000	£18,000	£12,000		£18,000

8. The table below contains cashflow information about a business. Calculate the **cash outflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£5,000	£8,000		£2,000	£7,000

9. The table below contains cashflow information about a business. Calculate the **cash inflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£12,000		£9,000	£7,000	£19,000

10. The table below contains cashflow information about a business. Calculate the **net cashflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£10,000	£15,000	£8,000		£17,000

Section 13 score: /10

Key Things to Remember About Net Cashflow

Write down any key information that you need to remember about average rate of return in this box.

Section 14: Opening and Closing Balances (Cashflow)

1. A small bakery has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	March (£)	April (£)	May (£)
Opening Balance	£5,000		
Cash Inflow	£7,000	£7,500	£7,200
Cash Outflow	£6,500	£7,000	£6,800
Net Cash Flow	£500	£500	£400
Closing Balance			

2. A digital marketing agency has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	June (£)	July (£)	August (£)
Opening Balance	£12,000		
Cash Inflow	£15,000	£16,000	£15,500
Cash Outflow	£13,500	£14,000	£13,800
Net Cash Flow	£1,500	£2,000	£1,700
Closing Balance			

3. A clothing store has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	January (£)	February (£)	March (£)
Opening Balance	£20,000		
Cash Inflow	£22,000	£23,000	£22,500
Cash Outflow	£19,500	£20,000	£19,800
Net Cash Flow	£2,500	£3,000	£2,700
Closing Balance			

4. A retail store has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	March (£)	April (£)	May (£)
Opening Balance	£8,000		
Cash Inflow	£10,000	£10,000	£11,500
Cash Outflow	£8,500	£12,000	£9,800
Net Cash Flow	£1,500	(£2,000)	£1,700
Closing Balance			

5. A construction company has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	June (£)	July (£)	August (£)
Opening Balance	£12,000		
Cash Inflow	£15,000	£15,000	£15,500
Cash Outflow	£12,500	£11,000	£16,600
Net Cash Flow	£2,500	£4,000	(£1,100)
Closing Balance			

6. A software development company has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	January (£)	February (£)	March (£)
Opening Balance	£18,000		
Cash Inflow	£21,000	£22,000	£18,300
Cash Outflow	£19,500	£21,000	£20,700
Net Cash Flow	£1,500	£1,000	(£2,400)
Closing Balance			

7. A restaurant has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	July (£)	August (£)	September (£)
Opening Balance	£6,000		
Cash Inflow	£8,000	£8,500	£8,200
Cash Outflow	£6,500	£7,000	£6,800
Net Cash Flow	£1,500	£1,500	£1,400
Closing Balance			

8. A consulting firm has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	October (£)	November (£)	December (£)
Opening Balance	£10,000		
Cash Inflow	£12,000	£13,000	£12,500
Cash Outflow	£11,200	£12,000	£11,800
Net Cash Flow	£800	£1,000	£700
Closing Balance			

9. A car repair shop has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	April (£)	May (£)	June (£)
Opening Balance	£5,000		
Cash Inflow	£5,500	£6,500	£6,200
Cash Outflow	£6,000	£6,000	£5,800
Net Cash Flow	(£500)	£500	£400
Closing Balance			

10. A landscaping company has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	November (£)	December (£)	January (£)
Opening Balance	£8,000		
Cash Inflow	£10,000	£11,000	£10,500
Cash Outflow	£9,000	£10,000	£9,800
Net Cash Flow	£1,000	£1,000	£700
Closing Balance			

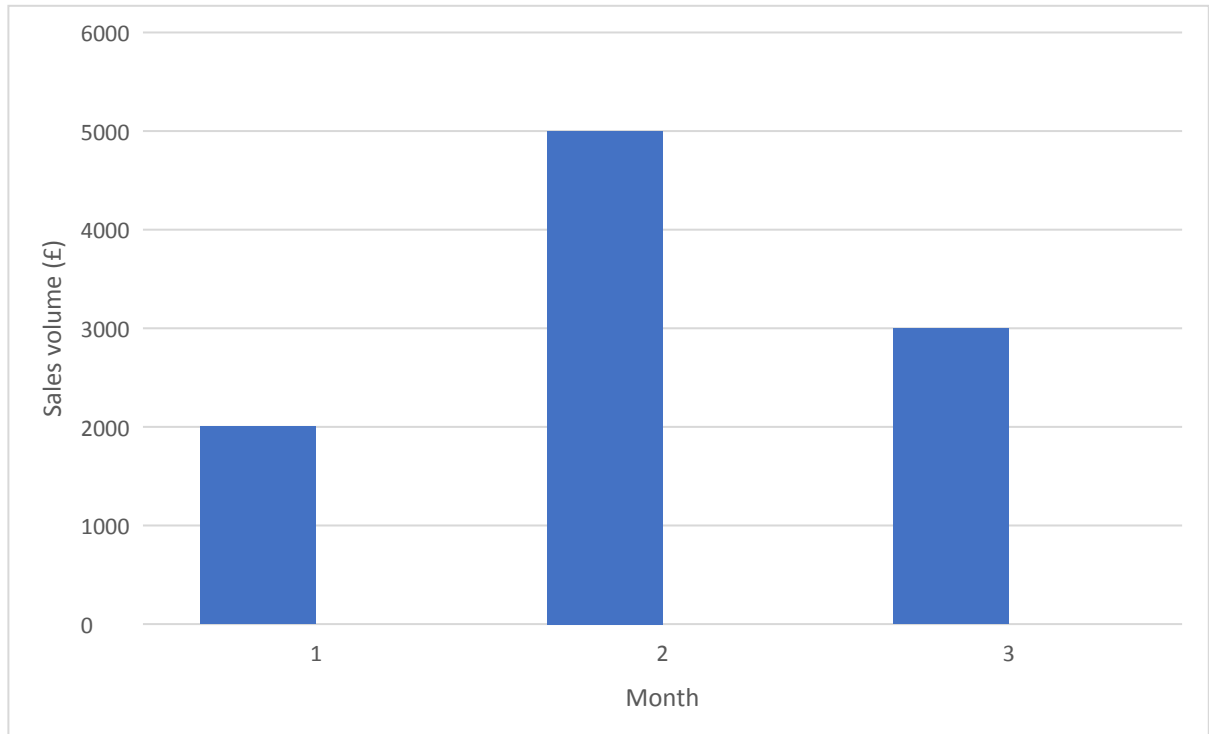
Section 14 score: /50

Key Things to Remember About Opening and Closing Balances

Write down any key information that you need to remember about opening and closing balances in this box.

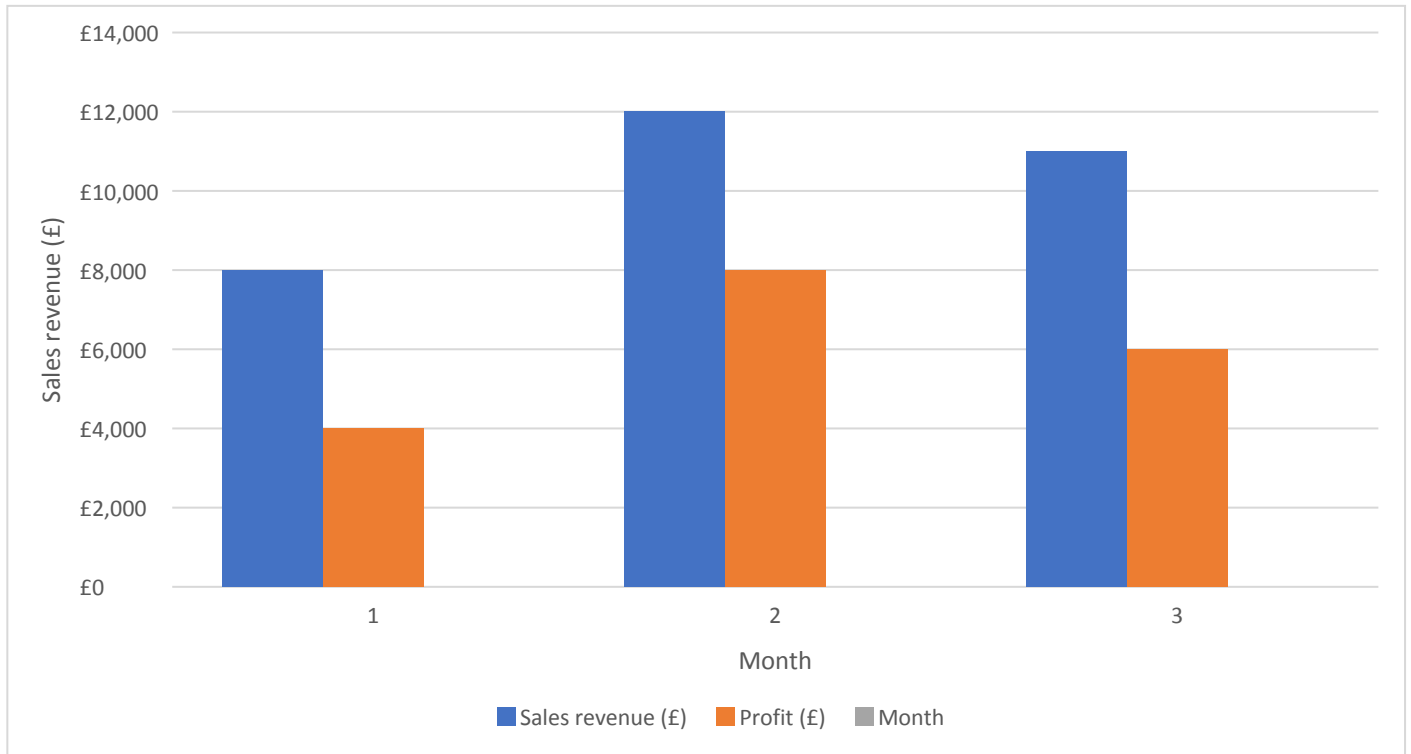
Section 15: Interpreting Information from Graphs and Charts

1. The bar chart below shows the sales volumes for a business over a three-month period for a product which has a selling price of £12.



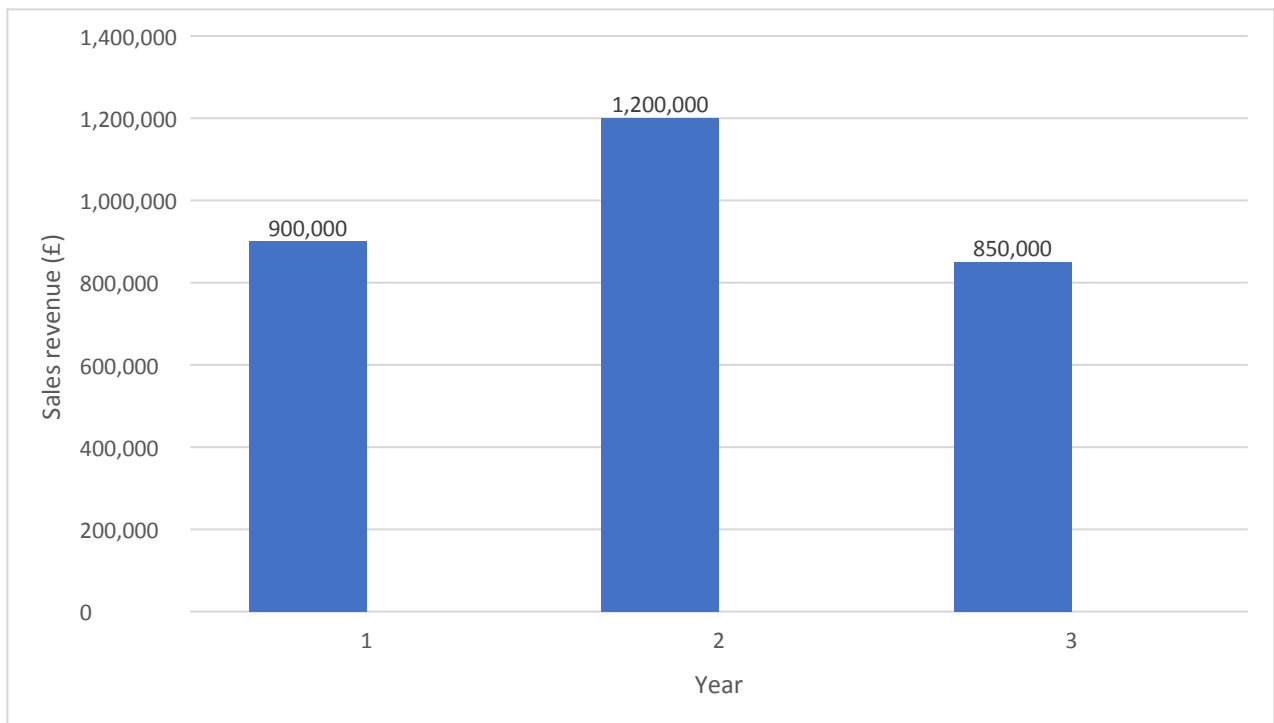
Using the information in the bar chart, calculate the sales revenue of the business for this three-month period. You are advised to show your workings (2)

2. The bar chart below shows information about the financial performance of a business over a three-month period.



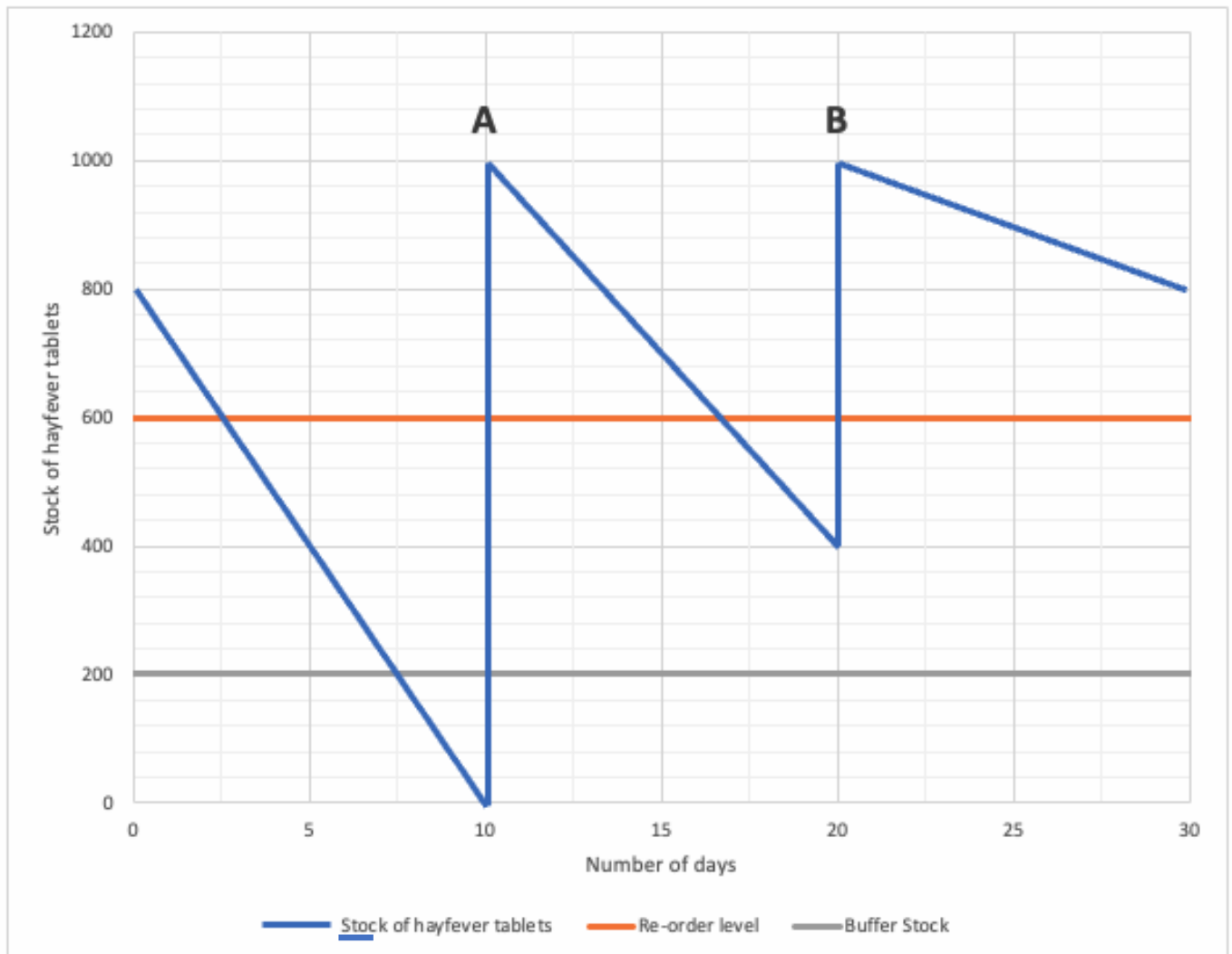
Using the information in the bar chart, calculate the total profit for this three-month period. You are advised to show your workings (2)

3. The bar chart below shows the sales revenue of a business over a three-year period.



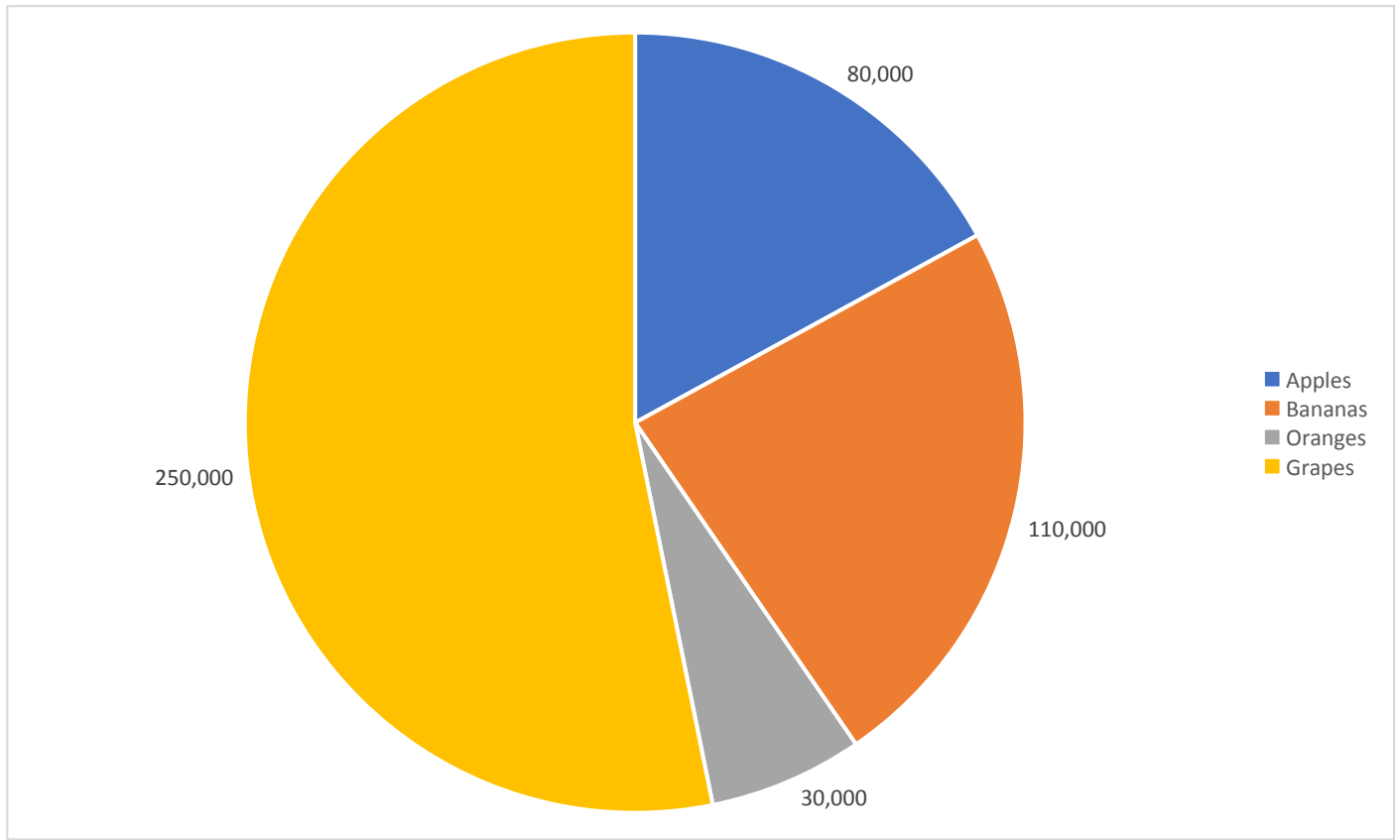
Using the information in the bar chart, calculate the percentage increase in sales revenue from year 1 to year 2. You are advised to show your workings (2)

4. The bar gate stock graph below shows the delivery details of hay fever tablets for a local pharmacy during the month of April. During this time it received two deliveries of hayfever tablets. These are marked as A and B.



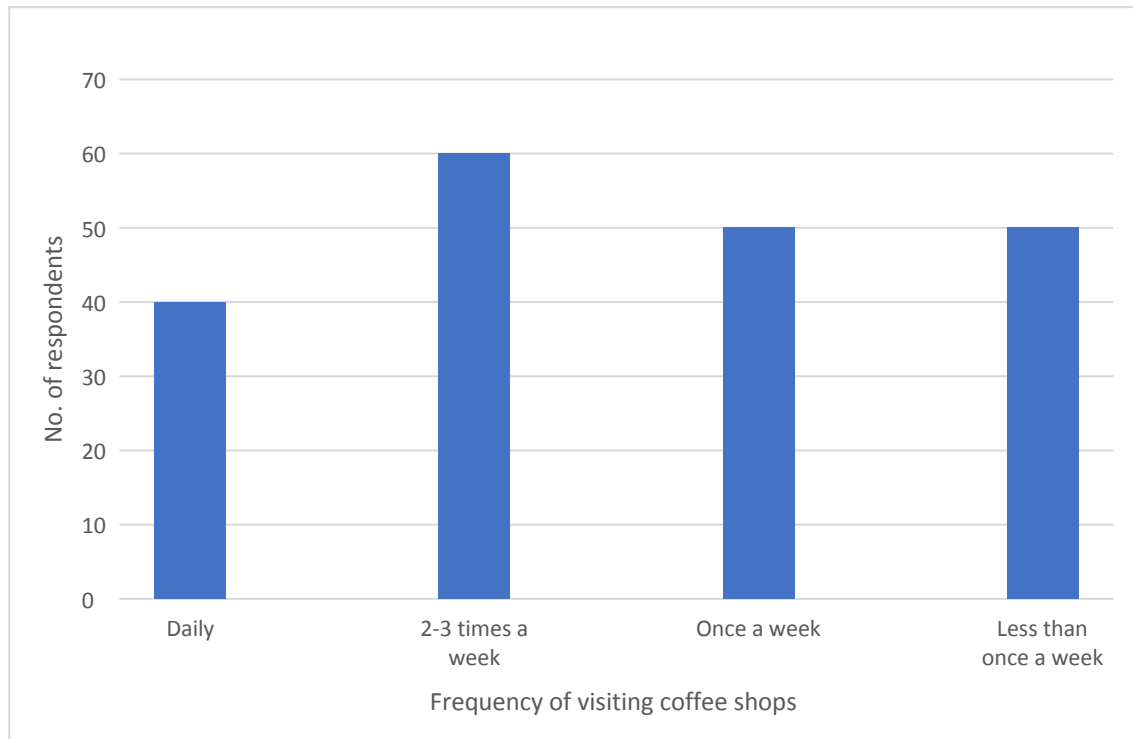
Using the information in the bar gate stock graph, Identify the re-order level and calculate the size of order A. You are advised to show your workings (2)

5. The pie chart below shows the total sales revenue made by four different product types (Apples, Bananas, Oranges, and Grapes) for a farm shop in a one-year period.



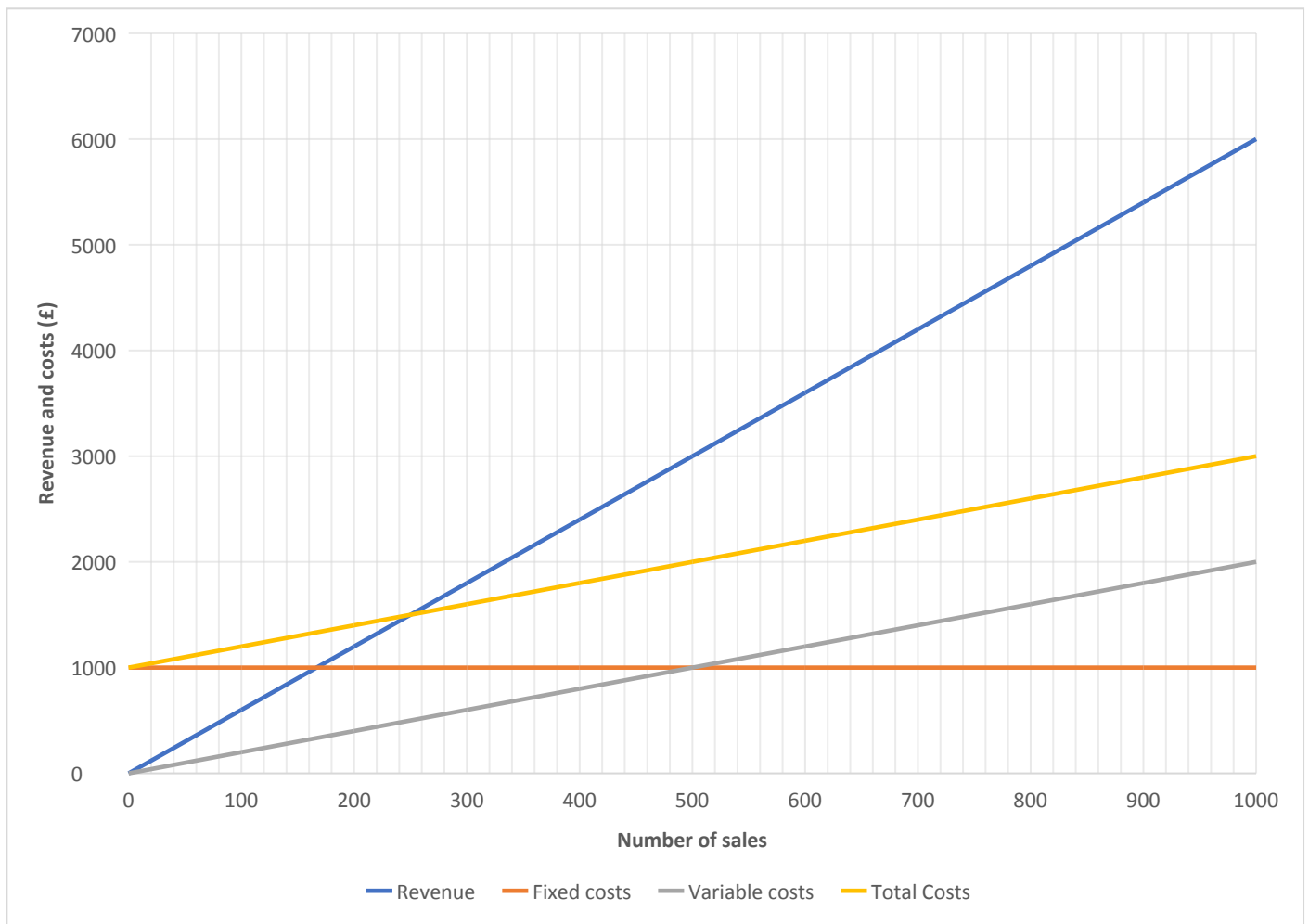
Using the information in the pie chart, calculate to 2 decimal places, the percentage of total sales revenue made by sales of Apples during this one-year period. You are advised to show your workings (2)

6. The bar chart below shows the results of a survey conducted by a coffee shop owner who asked participants how often they visited coffee shops.



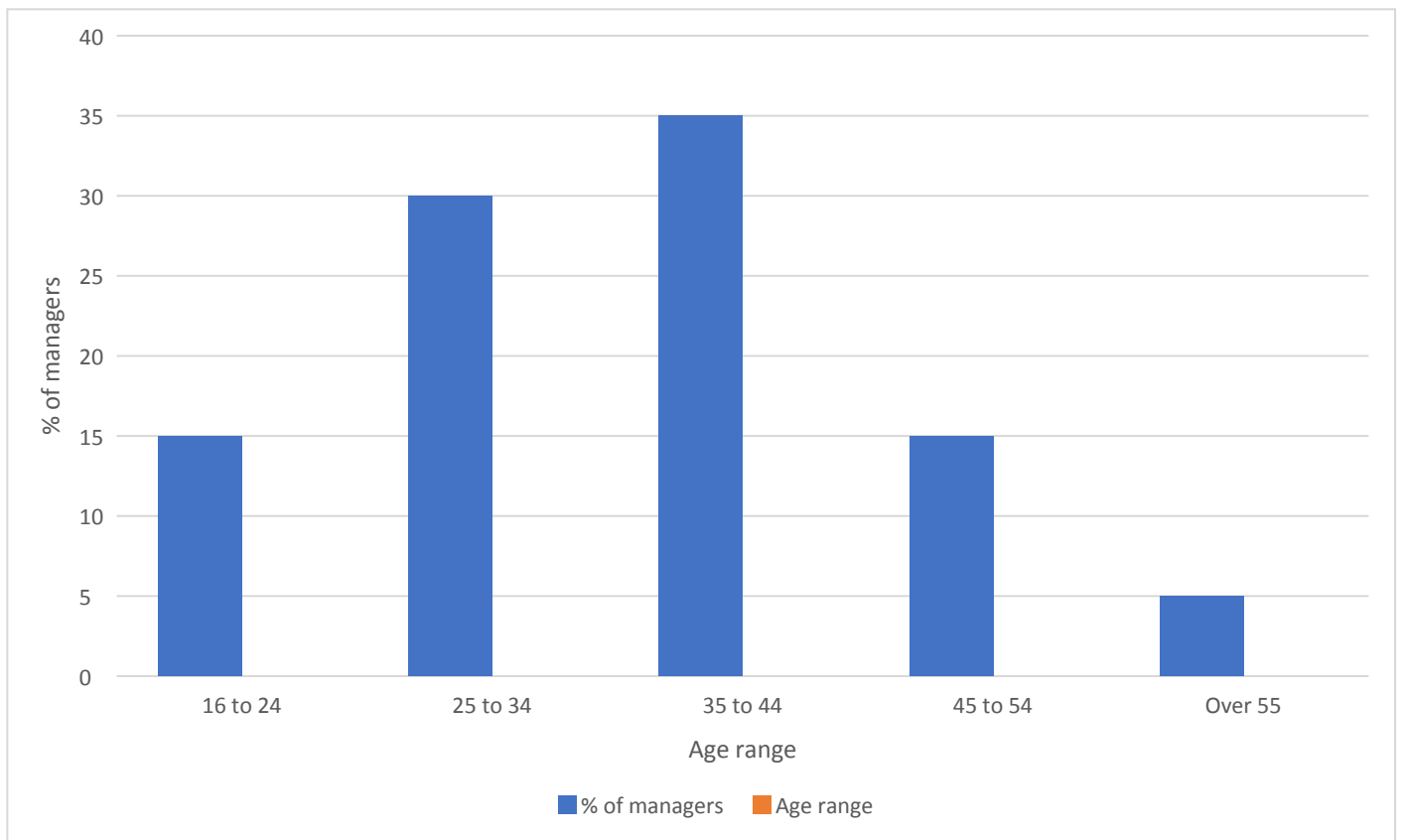
Using the information in the bar chart, calculate the percentage of participants who visited coffee shops at least twice a week. You are advised to show your workings (2)

7. A business has produced the following break-even diagram, showing their costs and revenue.



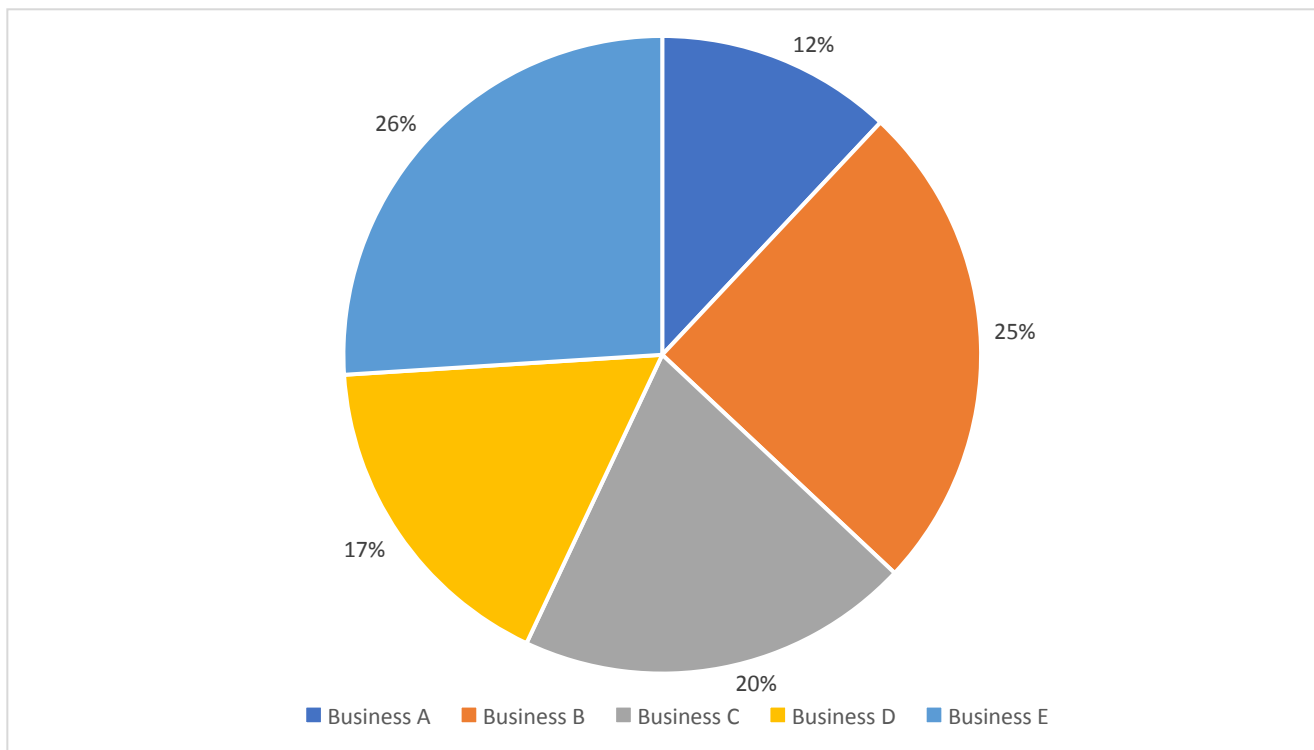
Using the information in the break-even diagram, identify the total number of sales the business needs to break even in units and how much the business expects to generate in sales revenue. You are advised to show your workings (2)

8. The bar chart below shows the age of managers for a nationwide retail chain.



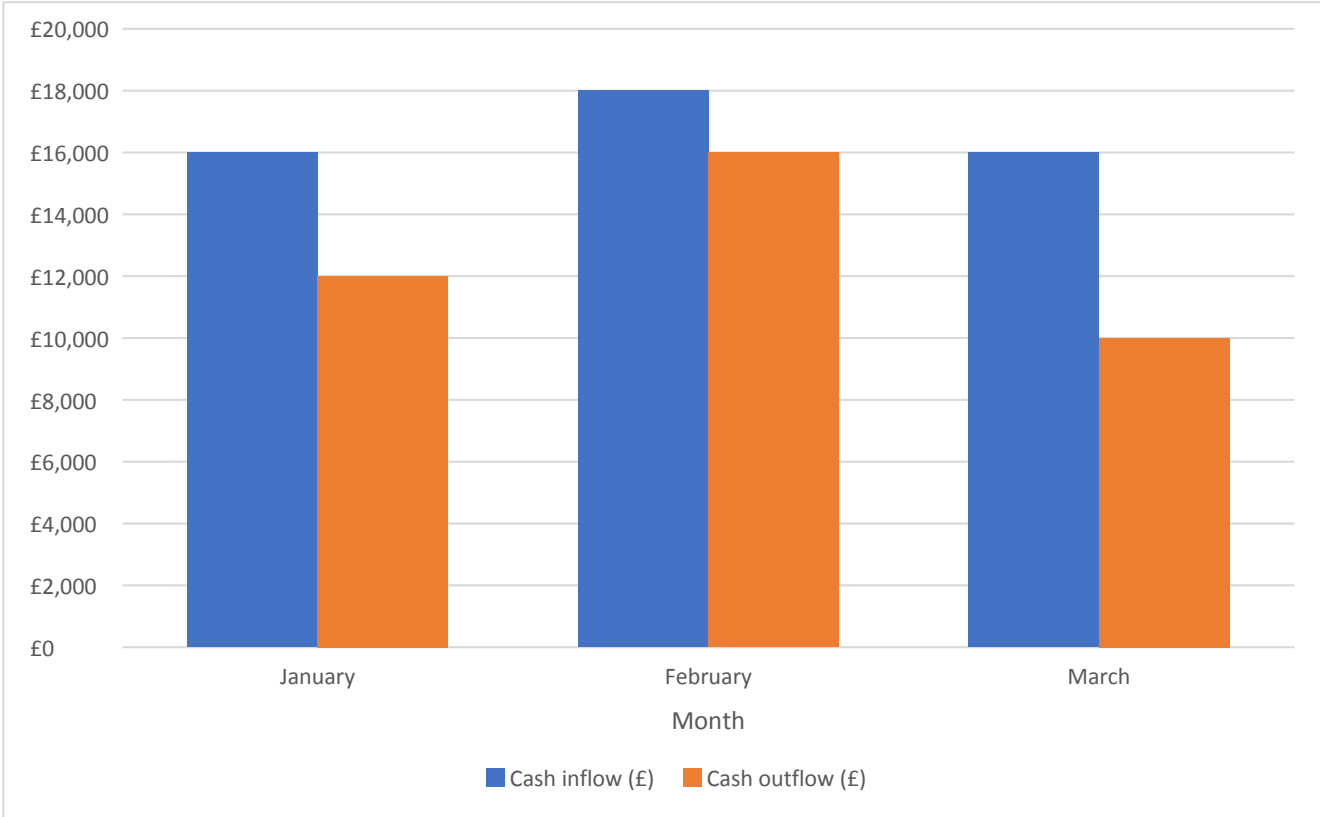
Using the information in the bar chart, identify what percentage of managers are under the age of 45. You are advised to show your workings (2)

9. The pie chart below shows the market share of each business in a market.



Using the information in the pie chart, calculate the market share of Business D. You are advised to show your workings (2)

10. The bar chart below shows information about the cash flow of a business over a three-month period.



Using the information in the bar chart, calculate which month has the best net cashflow over this three-month period. You are advised to show your workings (2)

Key Things to Remember About Interpreting Information from Graphs and Charts

Write down any key information that you need to remember about interpreting information from graphs and charts in this box.

Section 16: Break-even Level of Output

1. A small bakery sells cupcakes for £2 each. The fixed costs for the bakery amount to £500 per month, and the variable cost per cupcake is £1.50. Calculate the break-even level of output for the bakery. You are advised to show your workings (2)

2. A printing company charges £10 for each customised t-shirt. The fixed costs of the company amount to £2000, and the variable cost per t-shirt is £6. Calculate the break-even level of output for the printing company. You are advised to show your workings (2)

3. A fitness centre charges £30 per month for a membership. The fixed costs of the centre are £5000 per month, and the variable cost per member is £20. Calculate the break-even level of membership for the fitness centre. You are advised to show your workings (2)

4. A software company sells annual licenses for £1000 each. The fixed costs of the company are £10,000, and the variable cost per license is £200. Calculate the break-even point in revenue for the software company. You are advised to show your workings (2)

5. A coffee shop sells cups of coffee for £2.50 each. The fixed costs for the shop are £2000 per month, and the variable cost per cup of coffee is £1. Calculate the break-even level of output for the coffee shop. You are advised to show your workings (2)

6. A car wash charges £8 per car wash. The fixed costs for the car wash amount to £3000, and the variable cost per car wash is £5. Calculate the break-even level of output for the car wash. You are advised to show your workings (2)

7. A bookstore sells books for £20 each. The fixed costs for the bookstore are £4000, and the variable cost per book is £10. Calculate the break-even point in revenue for the bookstore. You are advised to show your workings (2)

8. A flower shop sells bouquets for £25 each. The fixed costs for the shop are £1500 per month, and the variable cost per bouquet is £15. Calculate the break-even level of output for the flower shop. You are advised to show your workings (2)

9. A restaurant charges £15 for each meal. The fixed costs for the restaurant amount to £5000 per month, and the variable cost per meal is £8. Calculate the break-even level of output for the restaurant. You are advised to show your workings (2)

10. A consulting firm charges £100 per hour for their services. The fixed costs of the firm are £10,000, and the variable cost per hour is £50. Calculate the break-even point in revenue for the consulting firm. You are advised to show your workings (2)

Section 16 score: /20

Key Things to Remember About Break-even Level of Output

Write down any key information that you need to remember about break-even level of output in this box.

Section 17: Margin of Safety

1. The break-even sales for a business are £300,000. The budgeted sales for the period are £400,000. Calculate the margin of safety. You are advised to show your workings (2)

2. The break-even sales for a business are £150,000. The actual sales for the period are £120,000. Calculate the margin of safety. You are advised to show your workings (2)

3. A software development company has fixed costs of £50,000 and a selling price per unit of £100. The variable cost per unit is £60. The actual sales for the period are 1,200 units. Calculate the margin of safety. You are advised to show your workings (2)

4. The break-even point in revenue for a business is £500,000 and the business has a margin of safety of £100,000. Calculate the actual sales for the period. You are advised to show your workings (2)

5. The actual sales for the period are £350,000 and the margin of safety is £50,000. Calculate break-even point in revenue. You are advised to show your workings (2)

6. A manufacturing company has fixed costs of £80,000 and a selling price per unit of £200. The variable cost per unit is £120. The actual sales for the period are 900 units. Calculate the margin of safety. You are advised to show your workings (2)

7. A retail store has fixed costs of £30,000 and a selling price per unit of £50. The variable cost per unit is £20. The actual sales for the period are 2,000 units. Calculate the margin of safety. You are advised to show your workings (2)

8. The break-even point in revenue for a business is £250,000 and the margin of safety is £135,000. Calculate the actual sales. You are advised to show your workings (2)

9. The break-even point in revenue for a business is £180,000. The actual sales for the period are £200,000. Calculate the margin of safety. You are advised to show your workings (2)

10. A restaurant has fixed costs of £40,000 and a selling price per meal of £20. The variable cost per meal is £10. The actual sales for the period are 3,500 meals. Calculate the margin of safety. You are advised to show your workings (2)

Section 17 score: /20

Key Things to Remember About Margin of Safety

Write down any key information that you need to remember about margin of safety in this box.

Section 18: Interest (on loans)

1. A business borrows £10,000 from a bank and repays a total of £11,500. Calculate the interest in percentage. You are advised to show your workings (2)

2. A business borrows £25,000 from a lender and repays a total of £29,500. Calculate the interest in percentage. You are advised to show your workings (2)

3. A business borrows £50,000 from a financial institution and repays a total of £55,000. Calculate the interest in percentage. You are advised to show your workings (2)

4. A business borrows £16,200 from a private investor and repays a total of £16,500. Calculate the interest in percentage. You are advised to show your workings (2)

5. A business borrows £40,000 from a venture capitalist and repays a total of £45,000. Calculate the interest in percentage. You are advised to show your workings (2)

6. A business borrows £64,000 from a peer-to-peer lending platform and repays a total of £69,200. Calculate the interest in percentage. You are advised to show your workings (2)

7. A business borrows £30,000 from a family member and repays a total of £31,500. Calculate the interest in percentage. You are advised to show your workings (2)

8. A business borrows £35,000 from a credit union and repays a total of £37,600. Calculate the interest in percentage. You are advised to show your workings (2)

9. A business borrows £26,000 from a government loan scheme and repays a total of £27,500. Calculate the interest in percentage. You are advised to show your workings (2)

10. A business borrows £50,000 from a commercial bank and repays a total of £55,500. Calculate the interest in percentage. You are advised to show your workings (2)

Section 18 score: /20

Key Things to Remember About Interest

Write down any key information that you need to remember about Interest in this box.

Section 19: Changes in Costs and Prices

1. A bakery is considering purchasing either Brand A flour at £2 per kilogram or Brand B flour at £1.80 per kilogram. The bakery requires 100 kilograms of flour. Calculate the price difference between the two options. You are advised to show your workings (2)

2. A clothing retailer currently sells a popular t-shirt at £15 per unit and sells 800 units per month. They are considering raising the selling price to £18 per unit. If they do raise the selling price, they expect the quantity sold to decrease to 700 units per month, calculate the difference in sales revenue between the current selling price and the potential new selling price. You are advised to show your workings (2)

3. A restaurant is deciding whether to hire a professional chef for £2,000 per month or a trainee chef for £1,200 per month. Calculate the monthly cost difference between the two options. You are advised to show your workings (2)

4. A manufacturing company is considering purchasing either Machine A for £10,000 or Machine B for £8,000. Calculate the cost difference between the two options. You are advised to show your workings (2)

5. A technology company currently sells a smartphone for £450 per unit and sells 500 units per month. They are considering reducing the selling price to £400 per unit. If they do reduce the selling price, they expect the quantity sold to increase to 600 units per month, calculate the difference in sales revenue between the current selling price and the potential new selling price. You are advised to show your workings (2)

6. A retailer is evaluating whether to source products locally for £5 each or import them from overseas for £3 each. The retailer needs 500 units. Calculate the price difference between the two options. You are advised to show your workings (2)

7. A coffee shop currently sells a specialty coffee for £3.50 per cup and sells 300 cups per day. They are considering increasing the selling price to £4.00 per cup. If they do increase the selling price, they expect the quantity sold to decrease to 250 cups per day, calculate the difference in sales revenue between the current selling price and the potential new selling price. You are advised to show your workings (2)

8. A construction company is deciding whether to use premium quality materials for £50,000 or standard quality materials for £40,000. Calculate the cost difference between the two options. You are advised to show your workings (2)

9. A bakery currently sells a loaf of bread for £2.50 and sells 200 loaves per day. They are considering reducing the selling price to £2.00. If they do reduce the selling price, they expect the quantity sold to increase to 250 loaves per day, calculate the difference in sales revenue between the current selling price and the potential new selling price. You are advised to show your workings (2)

10. A manufacturing company is considering purchasing raw materials from two different suppliers. Supplier A offers the raw materials at £5 per unit with a minimum order quantity of 100 units. Supplier B offers the same raw materials at £4.50 per unit with a minimum order quantity of 150 units. Calculate the difference in costs between the two suppliers and determine which option is more cost-effective. You are advised to show your workings (2)

Section 19 score: /20

Key Things to Remember About Changes in Costs and Prices

Write down any key information that you need to remember about changes in costs and prices in this box.

Section 20: Summary Quizzes

Summary Quiz 1

1. A coffee shop prepares 300 cups of coffee. The variable cost per cup is £0.30. Calculate the total variable cost. You are advised to show your workings (2)

2. A manufacturing company produces 500 units of a product. The production costs per unit are as follows: £10.50, £12.75, £9.80, £11.20, and £10.90. Calculate the average production cost per unit. You are advised to show your workings (2)

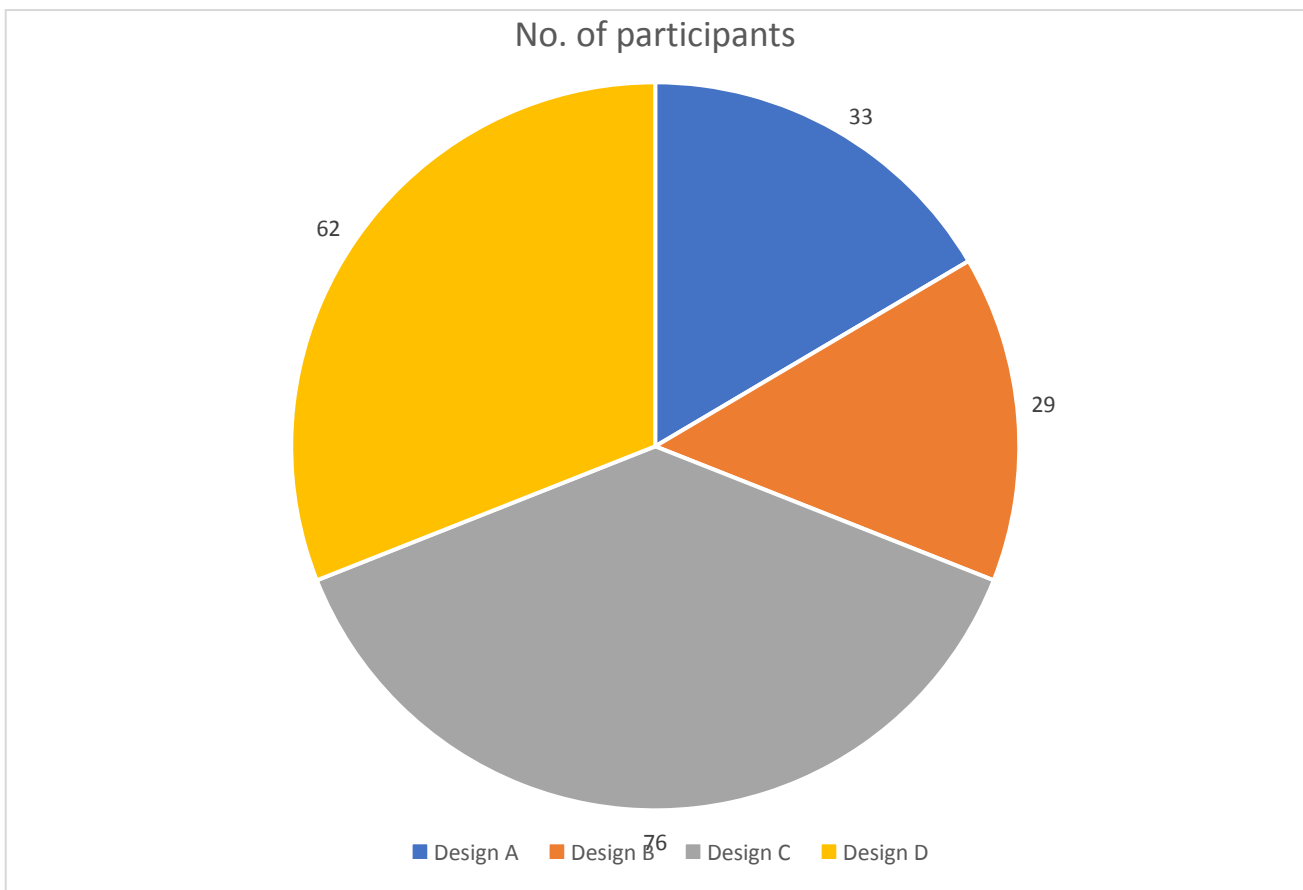
3. A toy store sells 200 toys at an average price of £20 each. The unit cost of purchasing each toy is £14. Calculate the gross profit margin. You are advised to show your workings (2)

4. A company sells mobile phone cases for £10 each. In the last month, they sold 500 cases. Calculate the sales revenue for the month. You are advised to show your workings (2)

5. A store had 100 customers last month, and it gained 20 new customers this month. Calculate the percentage increase in the customer base. You are advised to show your workings (2)

6. A manufacturing company produces and sells laptops. The fixed costs amount to £100,000 per month. The variable cost per laptop is £200, and the selling price is £800. Calculate the break-even level of output for the company. You are advised to show your workings (2)

7. The pie chart below shows the responses from the participants who took part in market research for a business to suggest which design they preferred.



Using the information in the pie chart, calculate to 2 decimal places, the percentage of participants who chose Design C. You are advised to show your workings (2)

8. A manufacturing company produces 500 units. The fixed costs are £10,000, and the variable cost per unit is £5. Calculate the total costs. You are advised to show your workings (2)

9. A coffee shop sells 300 cups of coffee at a price of £2.50 each. The total cost of preparing the coffee is £400. Calculate the profit. You are advised to show your workings (2)

10. A convenience store has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	April (£)	May (£)	June (£)
Opening Balance	£12,000		
Cash Inflow	£9,000	£8,200	£11,000
Cash Outflow	£8,000	£7,000	£14,400
Net Cash Flow	£1,000	£1,200	(£3,400)
Closing Balance			

11. Complete the table by calculating the missing sales, costs, or profit figures. (3)

	Month 1 (£)	Month 2 (£)	Month 3 (£)
Total Revenue		6,500	7,200
Fixed Costs	2,000	2,000	2,000
Variable Costs	1,500		1,800
Total Costs	3,500	3,500	3,800
Profit	1,500	3,000	

12. A retail store sells 500 units. The cost of goods sold per unit is £10, and the selling price per unit is £20. Calculate the gross profit. You are advised to show your workings (2)

13. A company is considering purchasing raw material A at £2 per unit or raw material B at £1.50 per unit. The company needs 500 units. Calculate the cost difference between the two options. You are advised to show your workings (2)

14. The table below contains information about a pharmacy. Calculate the net profit margin for the business. You are advised to show your workings (2)

Sales Revenue (£)	Gross Profit (£)	Other Operating Expenses (£)	Interest Expenses (£)
700,000	420,000	180,000	12,000

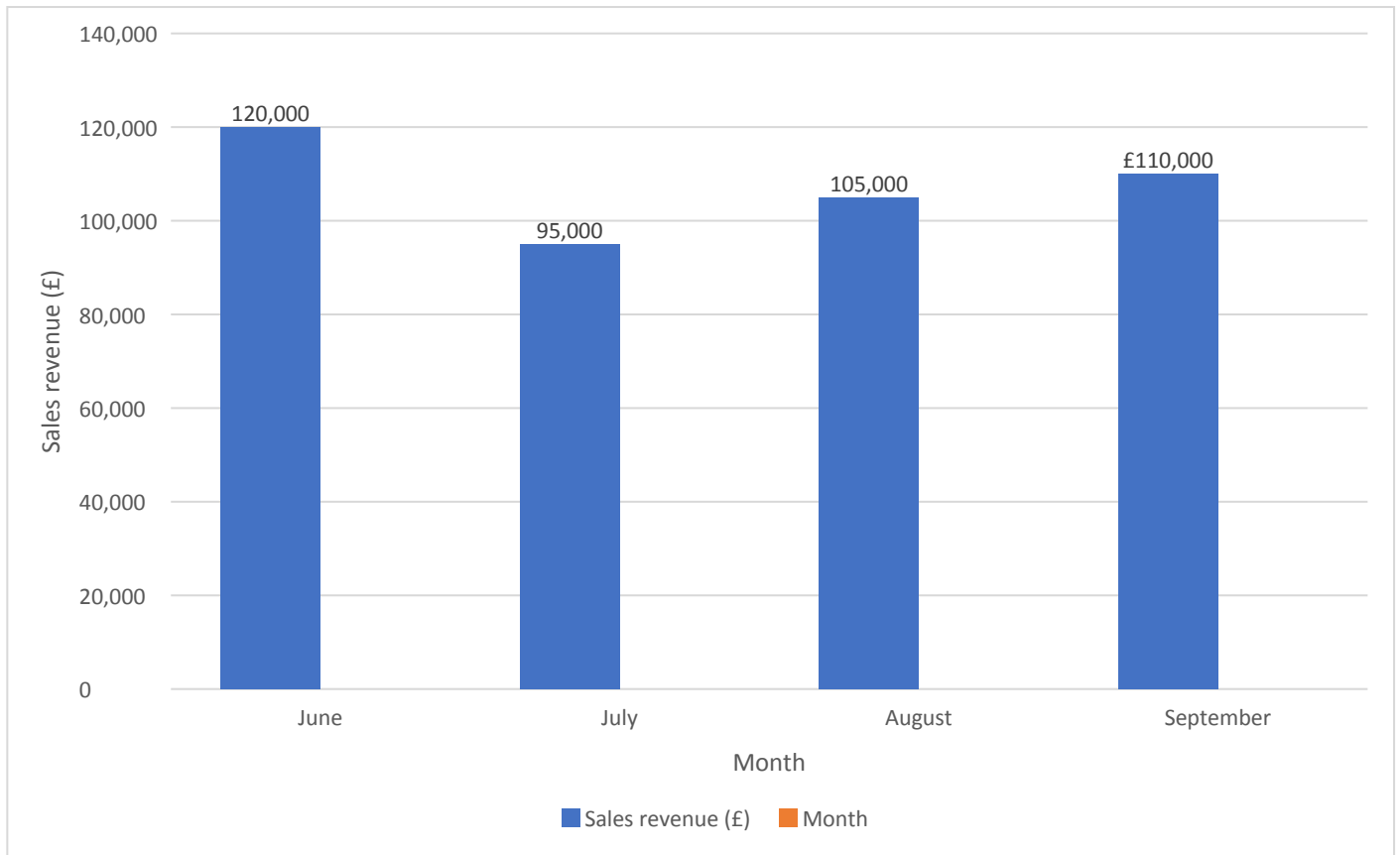
15. A software development company invests £200,000 in a new product and earns an average annual profit of £30,000. Calculate the average rate of return. You are advised to show your workings (2)

16. A consultancy firm charges a fixed fee of £5,000 per client engagement. The fixed costs for the firm amount to £20,000 per month, and the variable cost per client engagement is £1,000. Calculate the break-even level of client engagements for the consultancy firm. You are advised to show your workings (2)

17. A business borrows £12,000 from a bank and repays a total of £13,600. Calculate the interest in percentage. You are advised to show your workings (2)

18. A travel agency currently sells a holiday package for £1,500 per person and sells 80 packages per month. They are considering reducing the selling price to £1,400 per person. If they reduced the selling price, they expect the quantity sold to increase to 100 packages per month, calculate the difference in sales revenue between the current selling price and the potential new selling price. You are advised to show your workings (2)

19. A retail store generated sales revenue of £800,000 and achieved a gross profit of £500,000. With other operating expenses amounting to £200,000 and interest expenses of £10,000, calculate the net profit for the store. You are advised to show your workings (2)



20. The bar chart below shows the sales revenue of a business over a four-month period.

Using the information in the bar chart, calculate the average sales revenue the business made per month over the fourth-month period. You are advised to show your workings. You are advised to show your workings (2)

Summary quiz 1 score: /44

Summary Quiz 2

1. A coffee shop sells 300 cups of coffee at a price of £2.50 each. The unit cost of preparing each cup of coffee is £1. Calculate the gross profit margin. You are advised to show your workings (2)

2. A coffee shop sells coffee drinks. The fixed costs for the coffee shop amount to £7,500 per month. The variable cost per drink is £1.50, and the selling price is £3.50. Calculate the break-even level of drink sales for the coffee shop. You are advised to show your workings (2)

3. A company's stock price was £50 per share last month, and it decreased to £40 per share this month. Calculate the percentage decrease in the stock price. You are advised to show your workings (2)

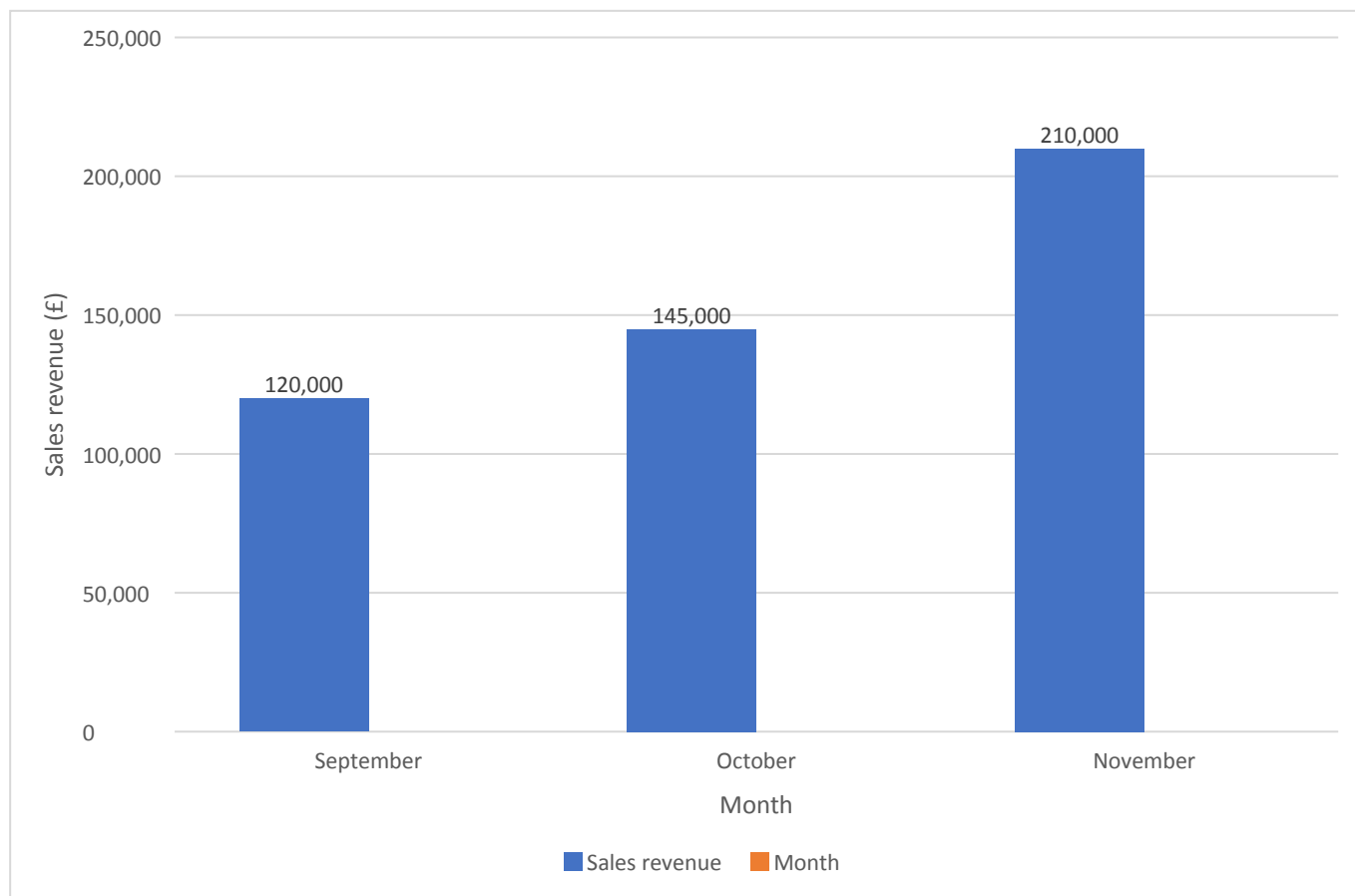
4. A car dealership sells luxury vehicles for £30,000 each. Last quarter, they sold 600 vehicles. Calculate the sales revenue for the quarter. You are advised to show your workings (2)

5. Complete the table by calculating the missing sales, costs, or profit figures. (5)

	Month 1 (£)	Month 2 (£)	Month 3 (£)
Total Revenue	7,500		8,500
Fixed Costs		3,000	3,000
Variable Costs	2,500	2,500	
Total Costs	5,500		5,800
Profit	2,000	2,500	

6. A restaurant prepares 200 meals. The cost of ingredients for each meal is £10, and the cost of packaging for each meal is £2. Calculate the total variable cost. You are advised to show your workings (2)

7. The bar chart below shows the sales revenue of a business over a three-year period.



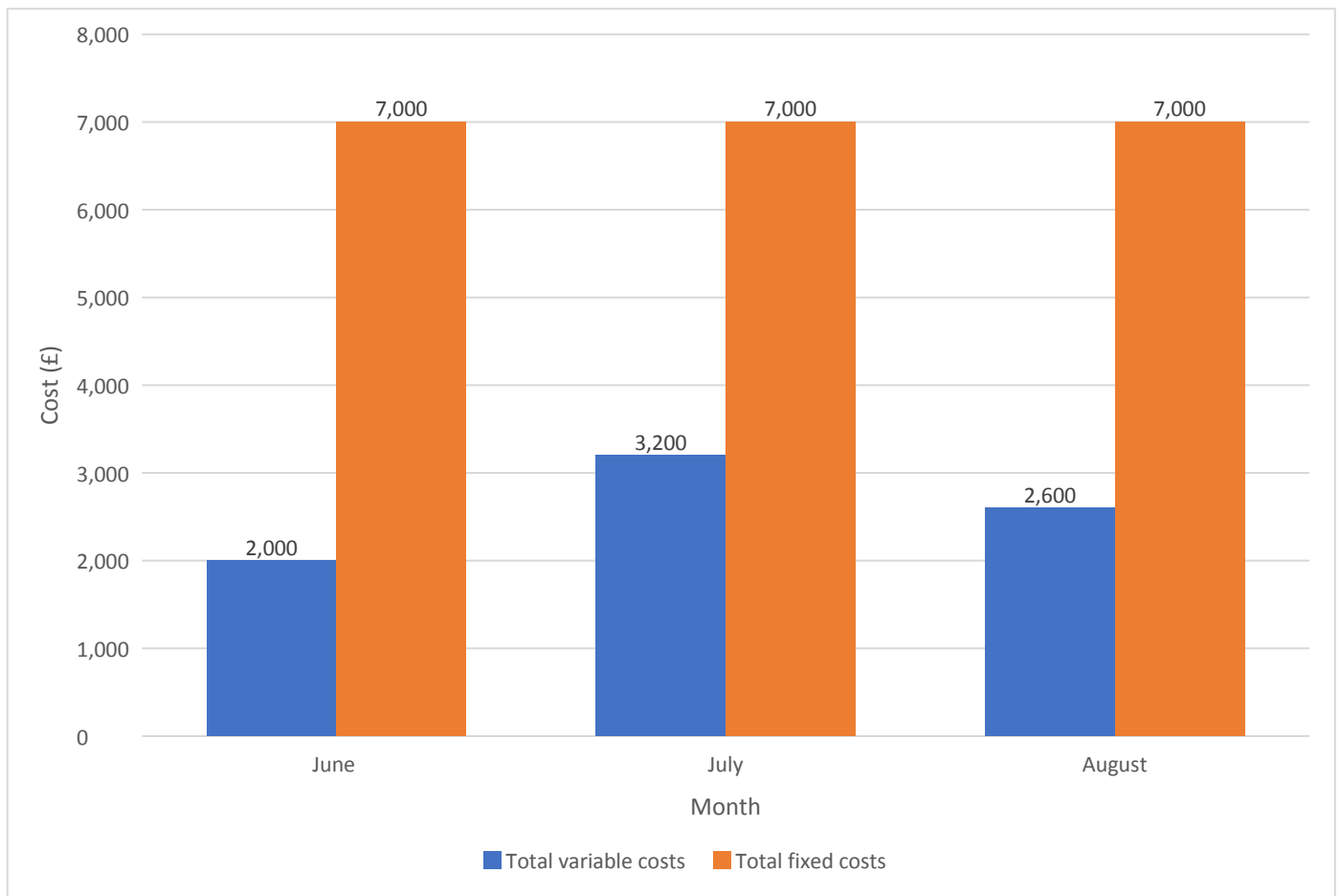
Using the information in the bar chart, calculate the percentage change in sales revenue from October to November. You are advised to show your workings (2)

8. A printing company prints 5,000 flyers. The cost of printing per flyer is £0.50, and the selling price per flyer is £1. Calculate the gross profit. You are advised to show your workings (2)

9. A bakery produces 400 loaves of bread. The fixed costs are £6,000, and the variable cost per loaf is £1.50. Calculate the total costs. You are advised to show your workings (2)

10. A restaurant serves 80 pizzas. The total cost of preparing the pizzas is £600. The average price per pizza is £12. Calculate the gross profit margin. You are advised to show your workings (2)

11. The bar chart below shows information about a business' costs over a three-month period.



Using the information in the bar chart, calculate the total variable costs over the three-month period. You are advised to show your workings (2)

12. A business borrows £35,000 from a credit union and repays a total of £39,200. Calculate the interest in percentage. You are advised to show your workings (2)

13. A fashion retailer generated sales revenue of £400,000 and achieved a gross profit of £250,000. With other operating expenses amounting to £100,000 and interest expenses of £4,000, calculate the net profit for the retailer. You are advised to show your workings (2)

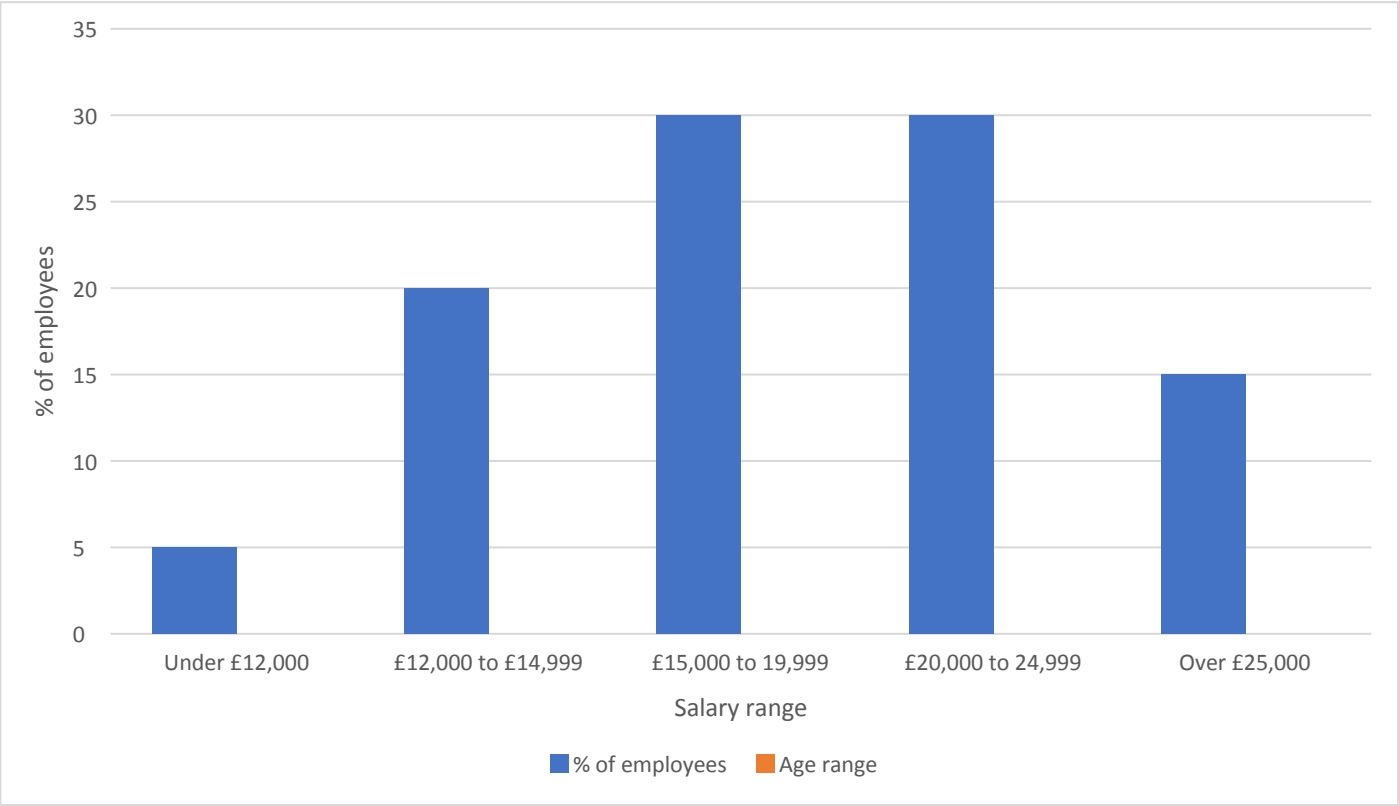
14. A construction company completes five projects. The total costs incurred for each project are as follows: £50,000, £45,000, £55,000, £52,000, and £48,000. Calculate the average cost of the projects. You are advised to show your workings (2)

15. A consulting firm has fixed costs of £60,000 and a selling price per hour of £100. The variable cost per hour is £60. The actual sales for the period are 1,800 hours. Calculate the margin of safety. You are advised to show your workings (2)

16. A construction company invests £200,000 in a new project. Over a period of 5 years, the project generates total profits of £120,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

17. A business borrows £28,000 from a government loan scheme and repays a total of £31,500. Calculate the interest in percentage. You are advised to show your workings (2)

18. The bar chart below shows the salary range for the employees of a supermarket.



Using the information in the bar chart, identify what percentage of employees earn a minimum of £15,000. You are advised to show your workings (2)

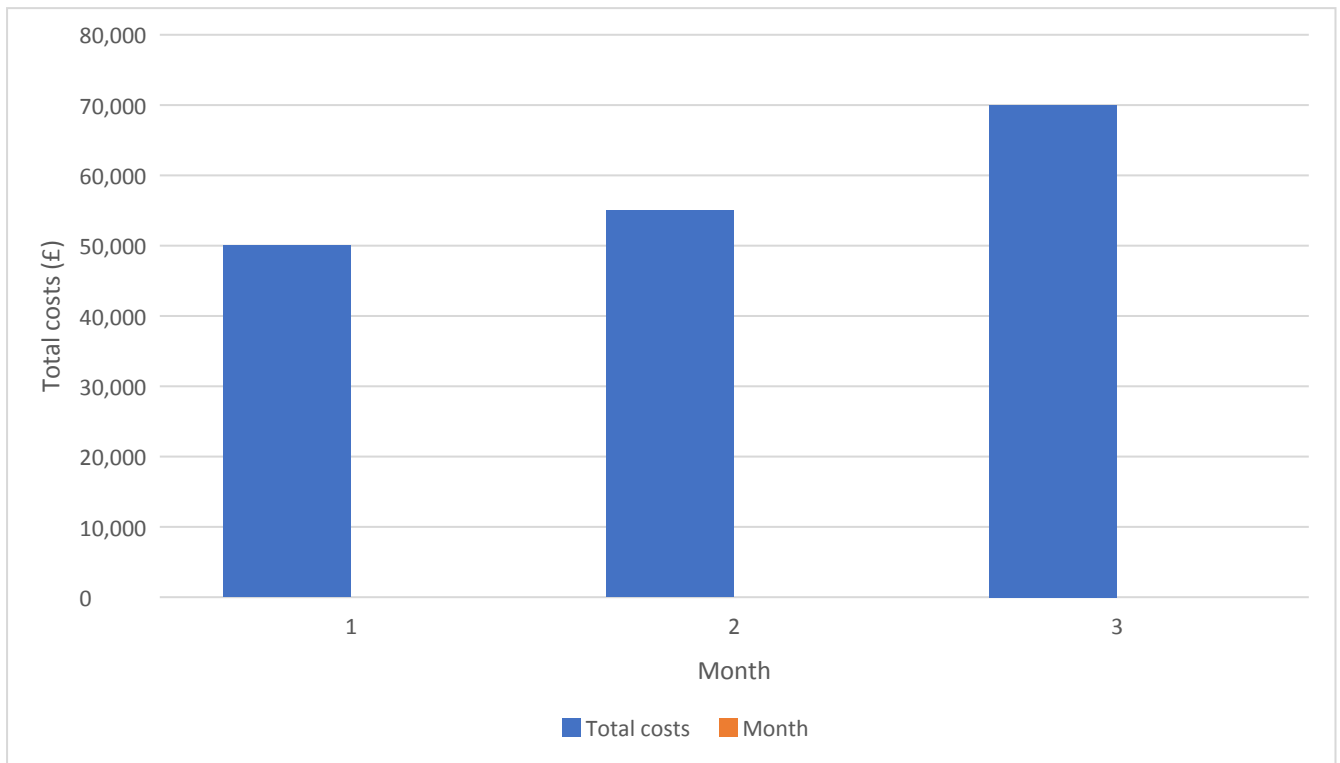
19. A restaurant is deciding between two suppliers for vegetables. Supplier A offers a bulk package of 200kg for £250, while Supplier B offers the same quantity for £300. However, Supplier A charges an additional delivery fee of £50, while Supplier B offers free delivery. The restaurant estimates using 150kg of vegetables. Calculate the cost difference between the two suppliers, including the delivery fee. You are advised to show your workings (2)

20. A PC repair shop has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	April (£)	May (£)	June (£)
Opening Balance	£9,000		
Cash Inflow	£7,500	£6,500	£8,000
Cash Outflow	£5,500	£8,000	£7,300
Net Cash Flow	£2,000	(£1,500)	£700
Closing Balance			

Summary Quiz 3

1. The bar chart below shows the total costs of a business over a three-year period.



Using the information in the bar chart, calculate the percentage change in total costs from month 1 to month 3. You are advised to show your workings (2)

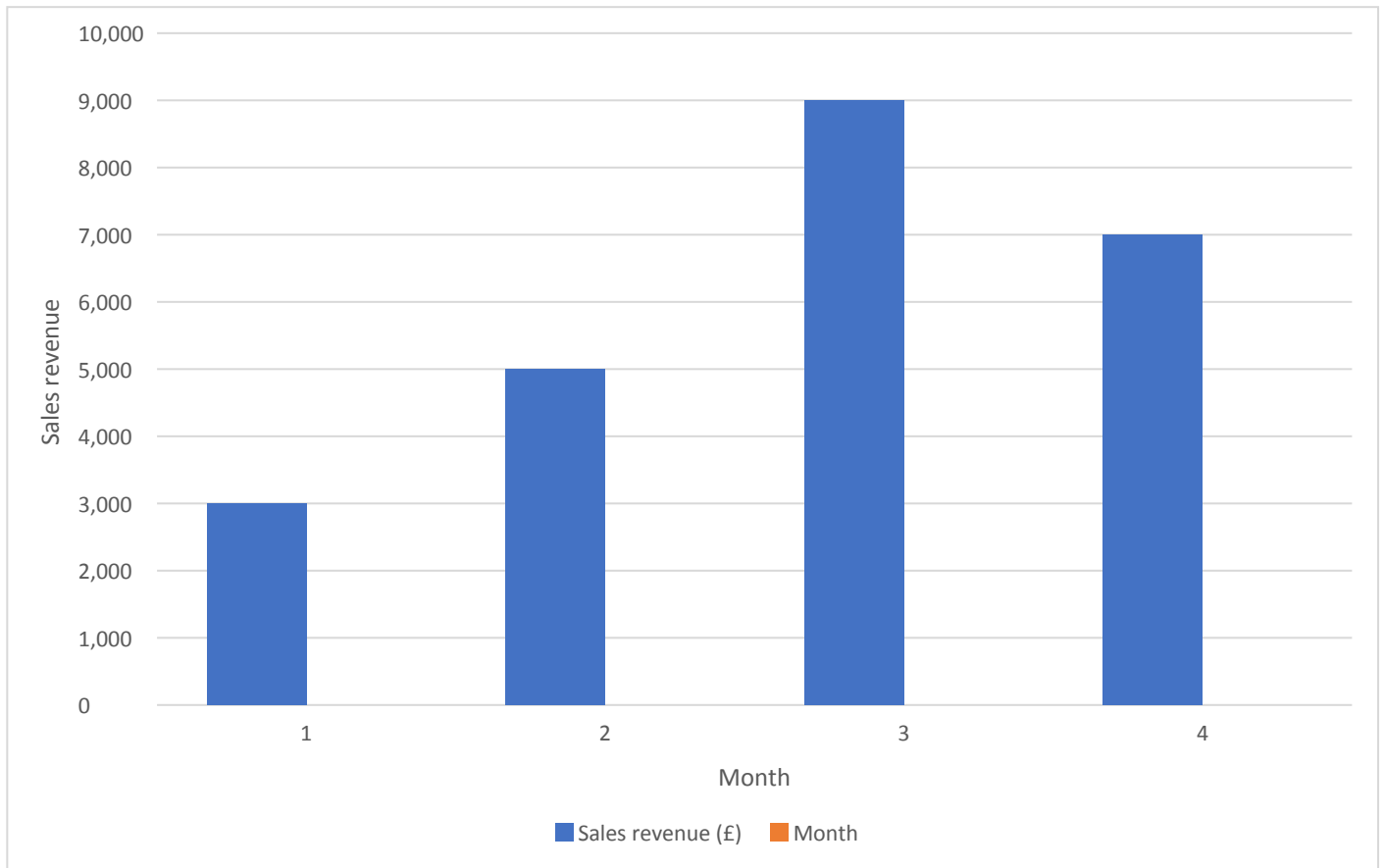
2. A construction company builds 10 houses. The fixed costs are £50,000, and the variable cost per house is £200,000. Calculate the total costs. You are advised to show your workings (2)

3. A retail store sells 200 products for £30 each and earns a gross profit of £2,600. Calculate the cost of goods sold. You are advised to show your workings (2)

4. The table below contains information about a fitness centre. Calculate the net profit margin for the business. You are advised to show your workings (2)

Sales Revenue (£)	Gross Profit (£)	Other Operating Expenses (£)	Interest Expenses (£)
400,000	280,000	120,000	9,000

5. The bar chart below shows the sales revenue of a business over a four-month period.



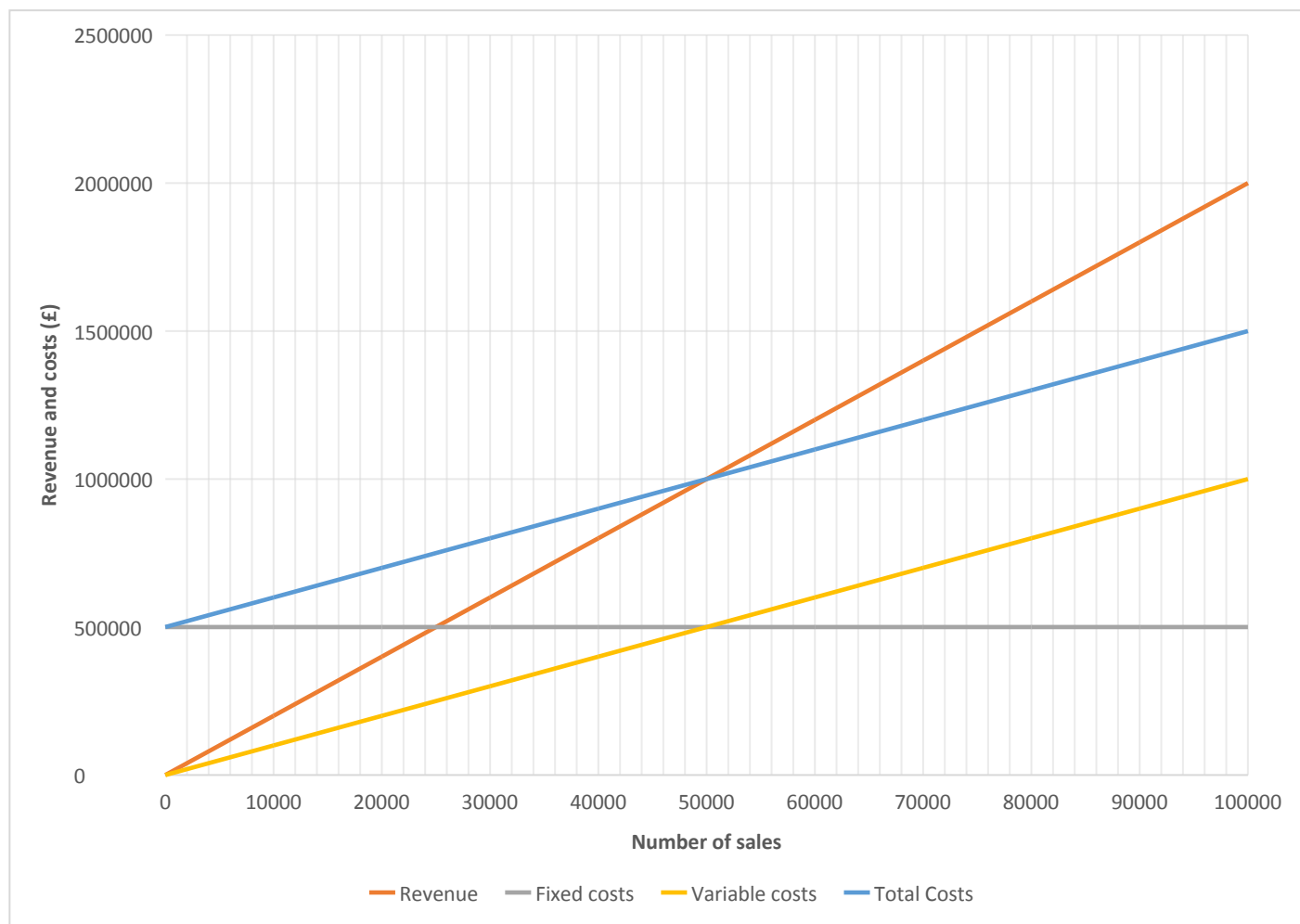
Using the information in the bar chart, calculate the average sales revenue the business made per month over the fourth-month period. You are advised to show your workings (2)

6. A printing company has fixed costs of £45,000 and a selling price per print job of £200. The variable cost per print job is £120. The actual sales for the period are 400 print jobs. Calculate the margin of safety. You are advised to show your workings (2)

7. A business borrows £26,300 from a venture capitalist and repays a total of £27,500. Calculate the interest in percentage. You are advised to show your workings (2)

8. A retail store sells four different brands of televisions. The wholesale prices for the televisions are as follows: £300, £350, £400, and £380. Calculate the average wholesale price of the televisions. You are advised to show your workings (2)

9. A business has produced the following break-even diagram, showing their costs and revenue.



Using the information in the break-even diagram, identify the total number of sales the business needs to break even. Then identify the profit made by the business by shading the correct region of the break-even diagram. You are advised to show your workings (2)

10. An online retailer sells handmade crafts for £18 each and generated a total of £4,500 in sales revenue in the past month. Calculate how many units the business sold. You are advised to show your workings (2)

11. A retail store invests £50,000 in inventory. Over a period of 3 years, the store generates total profits of £75,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

12. A clothing retailer operates multiple stores. The fixed costs for all stores combined amount to £200,000 per month. The variable cost per clothing item is £20, and the selling price is £80. Calculate the break-even point in revenue for the retailer. You are advised to show your workings (2)

13. A restaurant serves 80 pizzas. The variable cost per pizza is £4.50. Calculate the total variable cost. You are advised to show your workings (2)

14. A gym currently charges a monthly membership fee of £40 and has 150 members. They are considering increasing the membership fee to £45. If they do increase the membership fee, they expect the number of members to decrease to 120. Calculate the difference in sales revenue between the current membership fee and the potential new membership fee. You are advised to show your workings (2)

15. A restaurant serves 300 meals. The cost of ingredients per meal is £8, and the selling price per meal is £20. Calculate the gross profit. You are advised to show your workings (2)

16. Complete the table by calculating the missing sales, costs, or profit figures. (4)

	Month 1 (£)	Month 2 (£)	Month 3 (£)
Total Revenue	4,000		4,800
Fixed Costs		1,500	1,500
Variable Costs	1,000	1,200	1,300
Total Costs	2,500		2,800
Profit	1,500	1,800	

17. The Gourmet Restaurant had sales revenue of £400,000 and recorded a gross profit of £300,000. After deducting other operating expenses of £150,000 and interest expenses of £5,000, calculate the net profit for the restaurant. You are advised to show your workings (2)

18. A car repair shop has the following cash flow forecast for the next three months, complete the table by calculating the missing figures. (5)

Cash Flow Item	April (£)	May (£)	June (£)
Opening Balance	£7,000		£10,900
Cash Inflow	£8,000	£7,200	£7,000
Cash Outflow	£5,200		£9,800
Net Cash Flow		£1,100	(£2,800)
Closing Balance		£10,900	

19. A company had a 15% market share last year and it increased to a 18% market share this year. Calculate the percentage increase in the market share. You are advised to show your workings (2)

20. The table below contains information about a technology company. Calculate the gross profit margin for the company. You are advised to show your workings (2)

Business	Sales Revenue	Cost of Sales	Gross Profit
Technology	£510,000	£320,000	£190,000

Summary Quiz 4

1. A clothing store has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	January (£)	February (£)	March (£)
Opening Balance	£10,000		
Cash Inflow	£12,000	£13,000	£12,500
Cash Outflow	£9,500	£10,000	£9,800
Net Cash Flow	£2,500	£3,000	£2,700
Closing Balance			

2. A bakery invests £50,000 in new equipment. The equipment has a useful life of 10 years. Over this period, the bakery generates total profits of £180,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

3. An electronics retailer sells smartphones for £22 each. In the last month, they sold 450 smartphones. Calculate the sales revenue for the month. You are advised to show your workings (2)

4. A company had a 21% market share last year, and it increased to a 28% market share this year. Calculate the percentage increase in the market share. You are advised to show your workings (2)

5. A software company incurs monthly operating costs. The costs for the past five months are as follows: £15,000, £16,500, £14,800, £16,200, and £15,700. Calculate the average monthly operating cost. You are advised to show your workings (2)

6. A restaurant serves 400 meals. The fixed costs are £4,000, and the variable cost per meal is £5. The selling price per meal is £15. Calculate the profit. You are advised to show your workings (2)

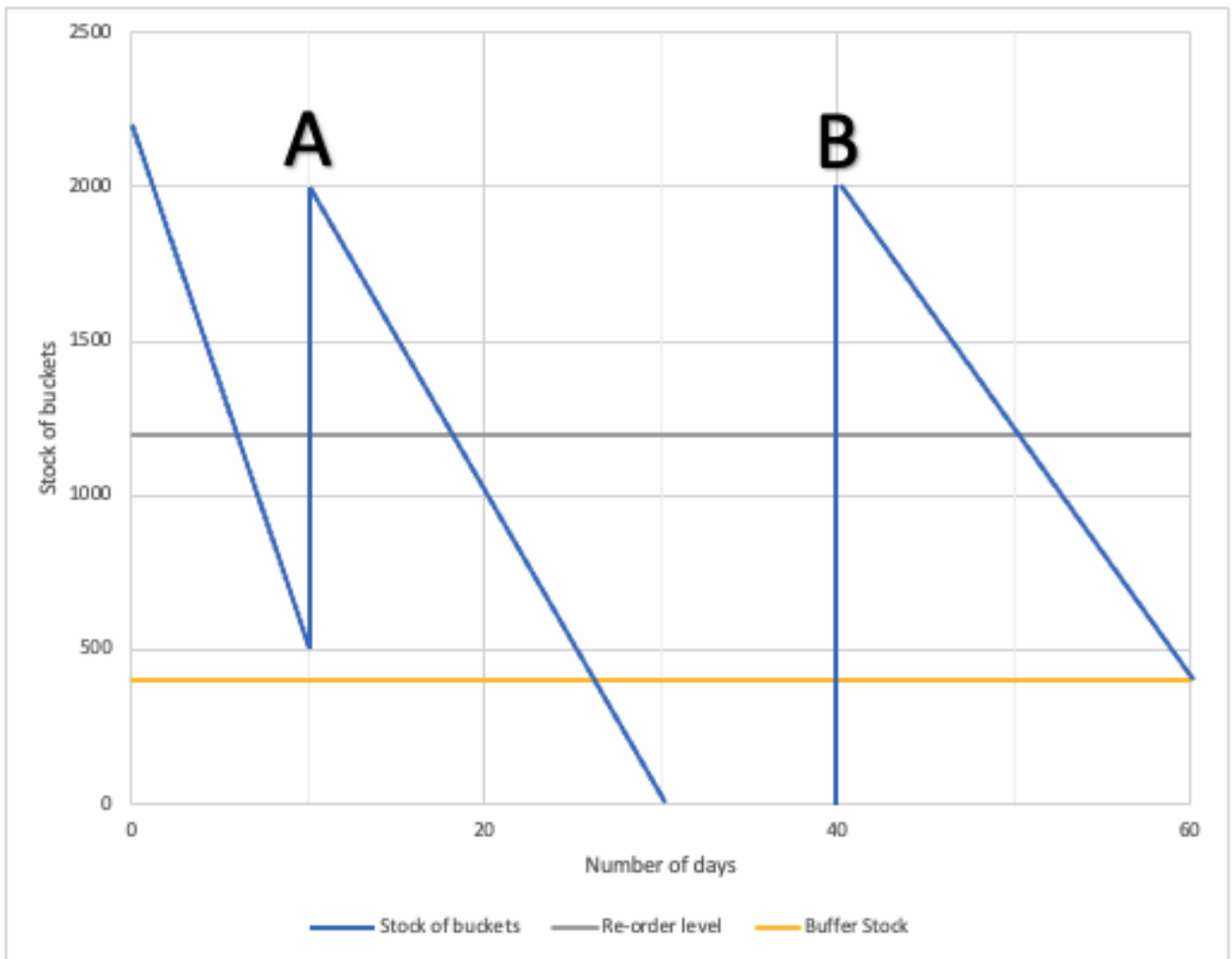
7. A bakery produces 400 loaves of bread. The variable cost per loaf is £1.50. Calculate the total variable cost. You are advised to show your workings (2)

8. A car manufacturer produces 150 cars. The fixed costs are £500,000, and the variable cost per car is £15,000. Calculate the total costs. You are advised to show your workings (2)

9. The table below contains information about a furniture store. Calculate the gross profit margin for the store. You are advised to show your workings (2)

Business	Sales Revenue	Cost of Sales	Gross Profit
Furniture	£620,000	£360,000	£260,000

10. The bar gate stock graph below shows the delivery details of buckets for a nationwide bucket specialist over a 60-day period during the summer. During this time it received two deliveries of buckets. These are marked as A and B.



Using the information in the bar gate stock graph, identify how long the business ran out buckets and calculate the size of order A. You are advised to show your workings. You are advised to show your workings (2)

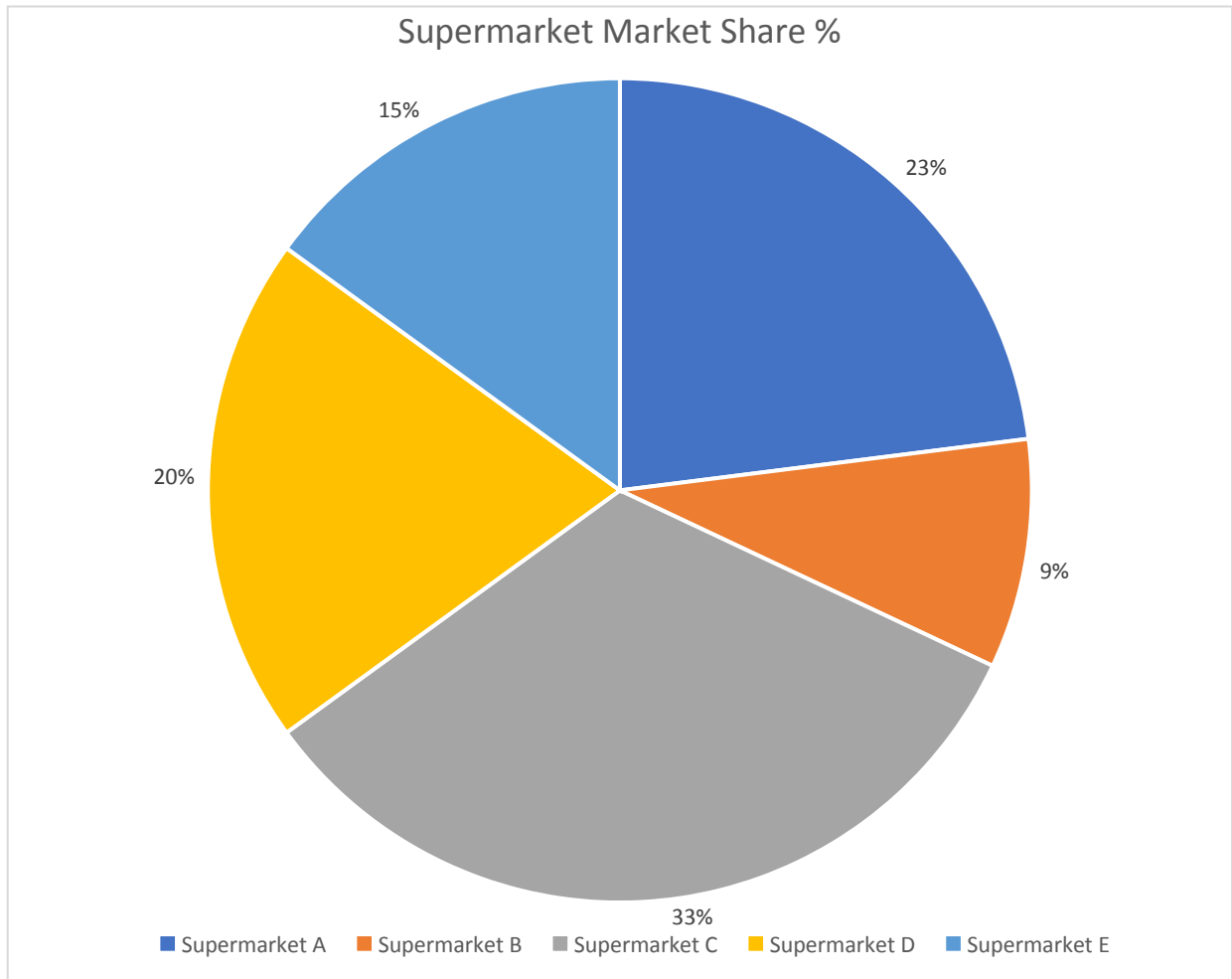
11. A car dealership sells 50 vehicles. The cost of vehicles per unit is £15,000, and the selling price per unit is £25,000. Calculate the gross profit. You are advised to show your workings (2)

12. An advertising agency had sales revenue of £200,000 and recorded a gross profit of £150,000. After deducting other operating expenses of £50,000 and interest expenses of £3,000, calculate the net profit for the agency. You are advised to show your workings (2)

13. The table below contains information about an advertising agency. Calculate the net profit margin for the business. You are advised to show your workings (2)

Sales Revenue (£)	Gross Profit (£)	Other Operating Expenses (£)	Interest Expenses (£)
600,000	420,000	180,000	15,000

14. The pie chart below shows the market share of the supermarket industry.



Using the information in the pie chart, calculate the market share of Supermarket B. You are advised to show your workings. You are advised to show your workings (2)

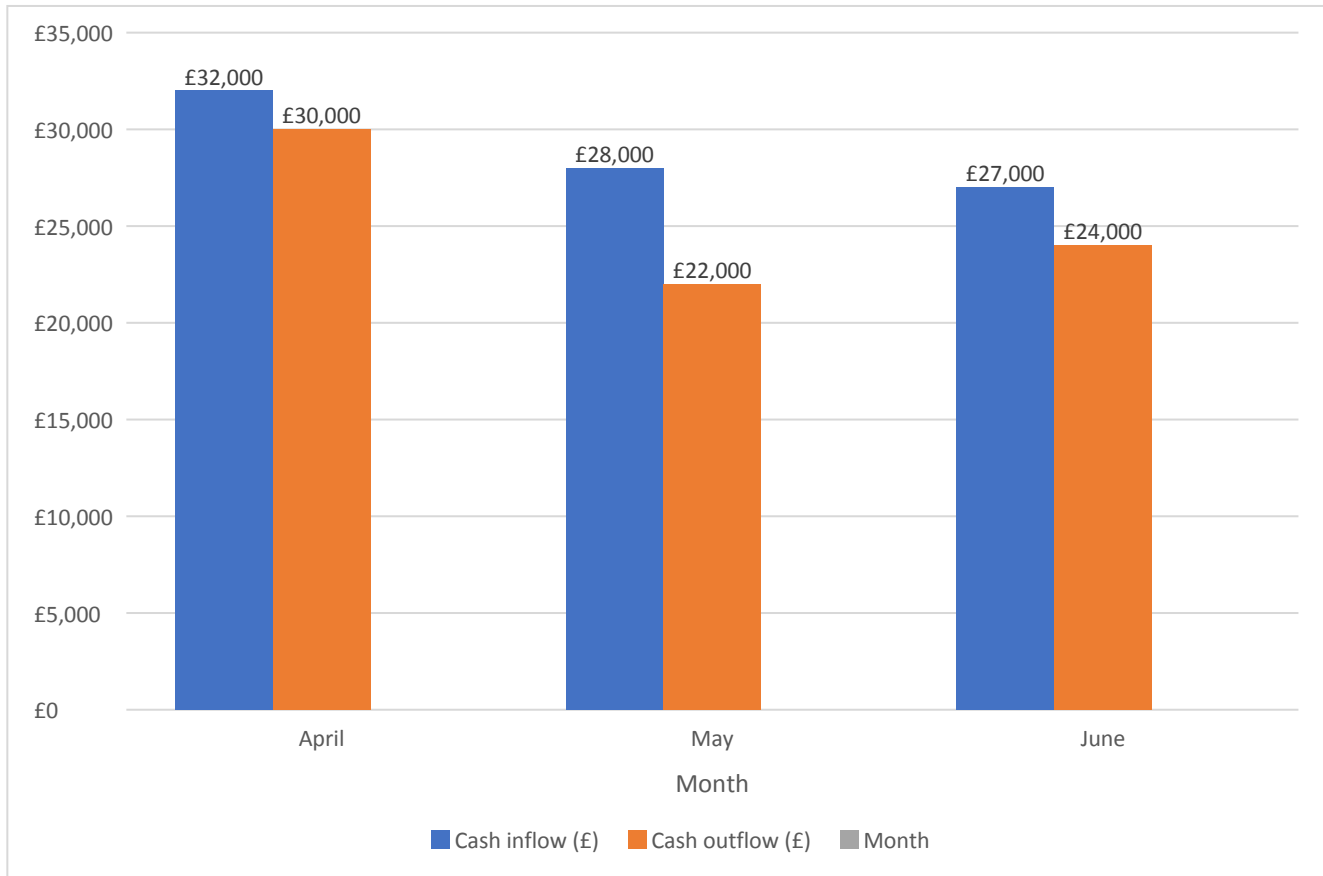
15. The table below contains cashflow information about a business. Calculate the **net cashflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£75,000	£56,000	£47,000		£84,000

16. A gym charges a monthly membership fee of £50. The fixed costs for the gym amount to £10,000 per month, and the variable cost per member is £20. Calculate the break-even level of memberships for the gym. You are advised to show your workings (2)

17. A business borrows £30,000 from a private investor and repays a total of £34,500. Calculate the interest in percentage. You are advised to show your workings (2)

18. The bar chart below shows information about the cash flow of a business over a three-month period.



Using the information in the bar chart, calculate which month has the best net cashflow over this three-month period. You are advised to show your workings. You are advised to show your workings (2)

19. A catering business has fixed costs of £60,000 and a selling price per event of £500. The variable cost per event is £300. The actual sales for the period are 100 events. Calculate the margin of safety. You are advised to show your workings (2)

20. A software company currently sells a license for £50 per user and has 500 users. They are considering increasing the price to £60 per user. If they do increase the price, they expect the number of users to decrease to 400, calculate the difference in sales revenue between the current price and the potential new price. You are advised to show your workings (2)

Summary Quiz 5

1. The table below contains information about a coffee shop. Calculate the net profit margin for the business. You are advised to show your workings (2)

Sales Revenue (£)	Gross Profit (£)	Other Operating Expenses (£)	Interest Expenses (£)
250,000	150,000	60,000	5,000

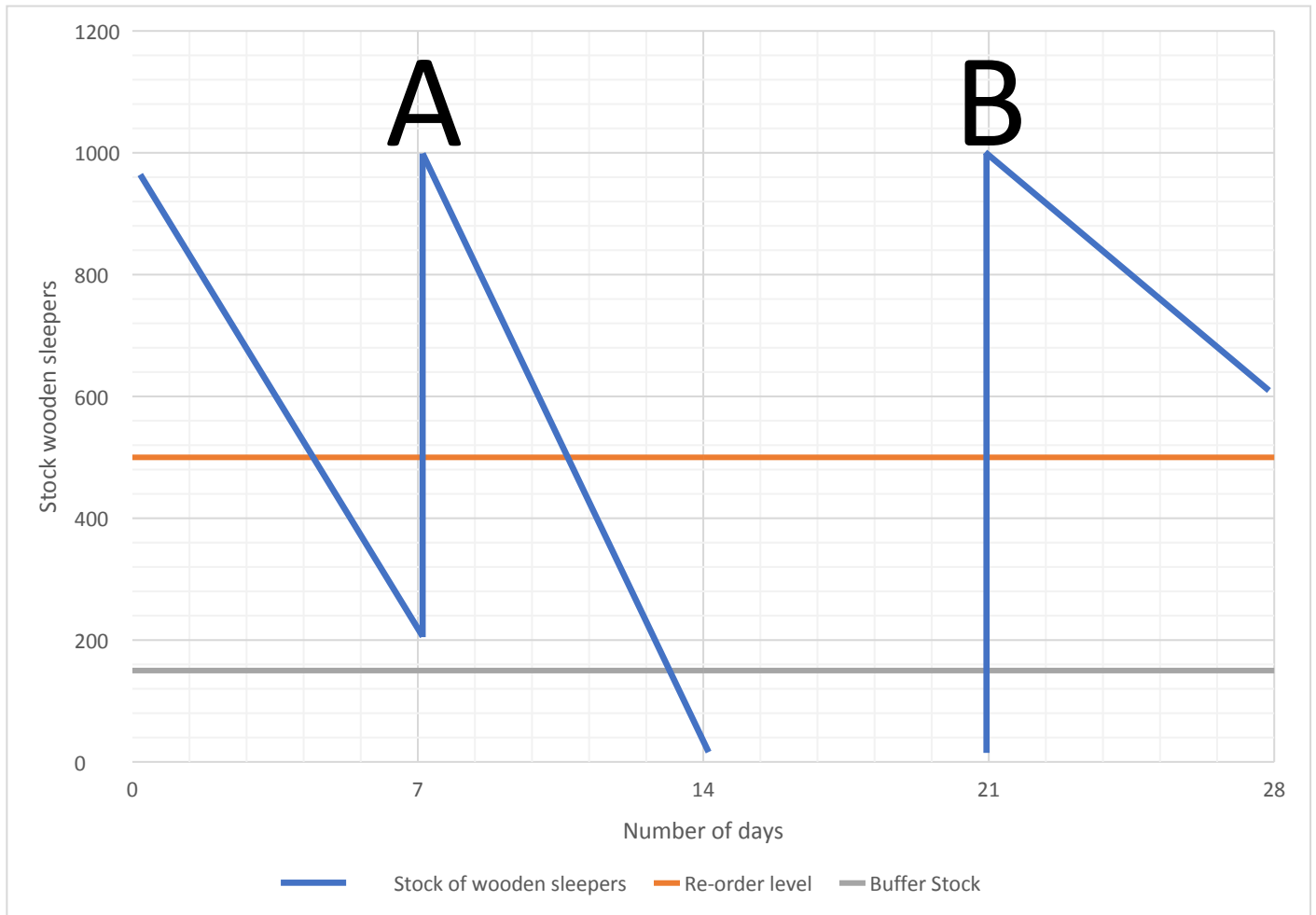
2. The table below contains cashflow information about a business. Calculate the **cash outflow** for the business. (1)

Opening Balance	Cash Inflow	Cash Outflow	Net Cashflow from Operations	Closing Balance
£15,000	£28,300		£6,500	£21,500

3. A retail store has the following cash flow forecast for the next three months, complete the table by calculating the opening and closing balances. (5)

Cash Flow Item	March (£)	April (£)	May (£)
Opening Balance	£21,000		
Cash Inflow	£11,000	£12,000	£11,500
Cash Outflow	£18,500	£10,000	£11,000
Net Cash Flow		£2,000	£500
Closing Balance			

4. The bar gate stock graph below shows the delivery details of wooden sleepers for a local garden centre over a 28-day period during the spring. During this time it received two deliveries of garden sleepers. These are marked as A and B.



Using the information in the bar gate stock graph, identify how long the business ran out garden sleepers during this 28-day period and calculate the size of order B. You are advised to show your workings (2)

5. Complete the table by calculating the missing sales, costs, or profit figures. (5)

	Month 1 (£)	Month 2 (£)	Month 3 (£)
Total Revenue		6,500	7,000
Fixed Costs	2,500		2,500
Variable Costs	2,000	2,000	
Total Costs		4,500	4,700
Profit	1,500		2,300

6. A toy store currently sells a board game for £20 and sells 100 units per week. They are considering reducing the selling price to £18. If they do reduce the selling price, they expect the quantity sold to increase to 120 units per week, calculate the difference in sales revenue between the current selling price and the potential new selling price. You are advised to show your workings (2)

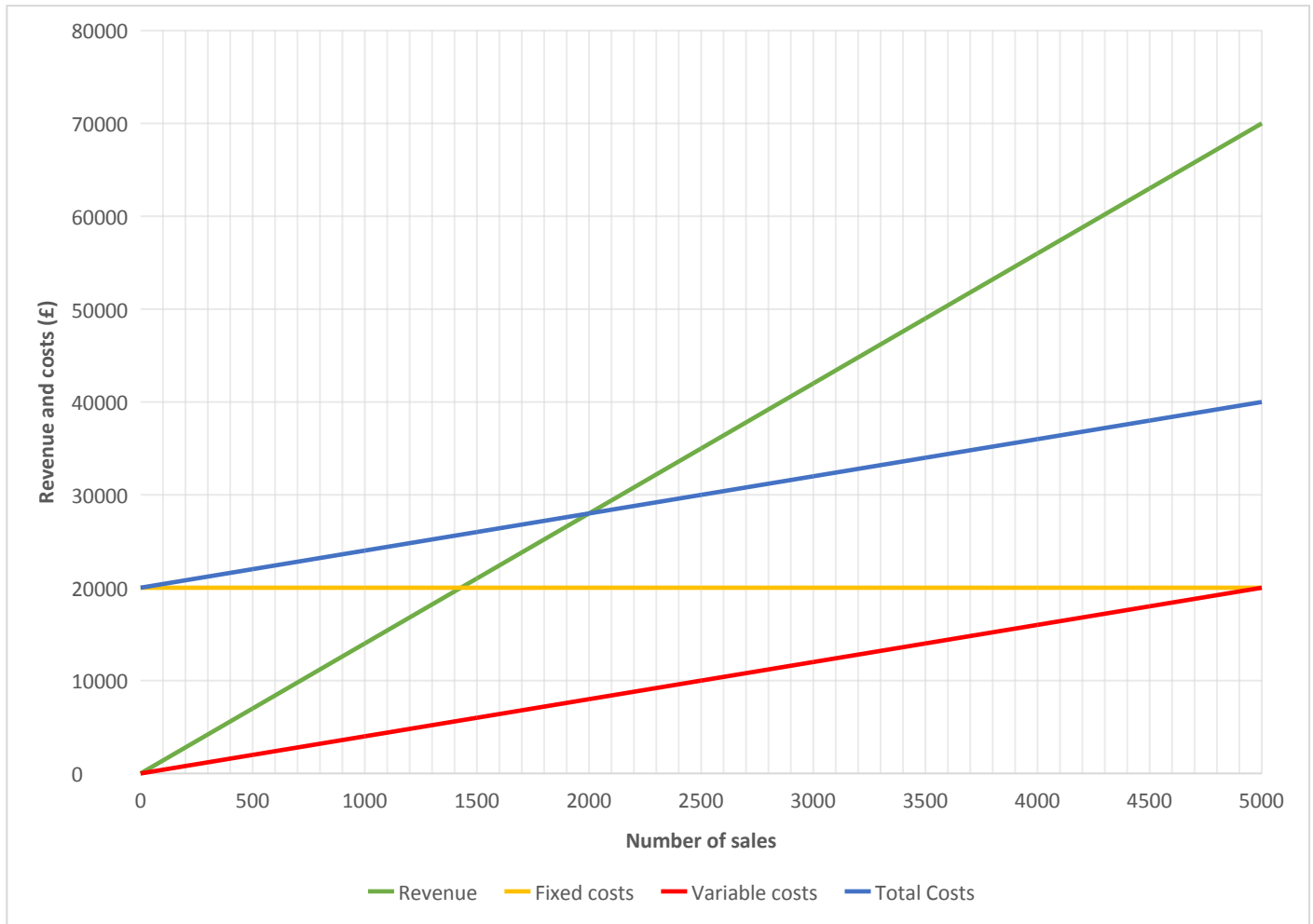
7. A business borrows £18,500 from a financial institution and repays a total of £21,000. Calculate the interest in percentage. You are advised to show your workings (2)

8. A bookstore sells books online. The fixed costs amount to £5,000 per month. The variable cost per book is £5 and the selling price is £15. Calculate the break-even level of book sales for the bookstore. You are advised to show your workings (2)

9. A software development company invests £300,000 in developing a new product. Over a period of 6 years, the product generates total profits of £400,000. Calculate the average rate of return as a percentage. You are advised to show your workings (2)

10. A construction company had sales revenue of £1,000,000 and recorded a gross profit of £600,000. After deducting other operating expenses of £250,000 and interest expenses of £12,000, calculate the net profit for the company. You are advised to show your workings (2)

11. A business has produced the following break-even diagram, showing their costs and revenue.



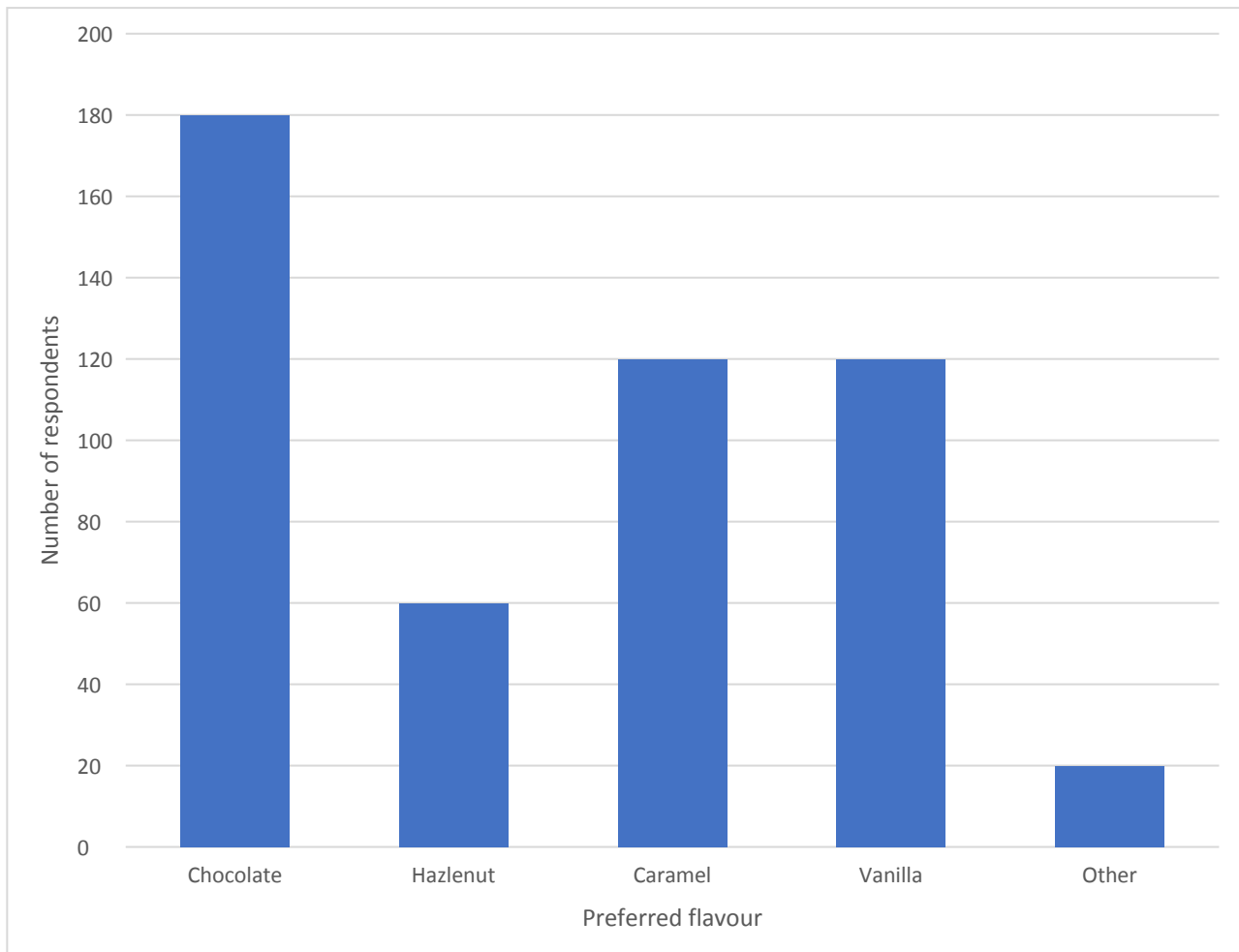
Using the information in the break-even diagram, identify the total number of sales the business needs to break even in units and how much the business expects to spend on variable costs if it sells 5,000 units. You are advised to show your workings (2)

12. A bakery sells 200 cakes. The cost of ingredients per cake is £7, and the selling price per cake is £16. Calculate the gross profit margin. You are advised to show your workings (2)

13. A toy company produces 250 stuffed animals. The variable cost per stuffed animal is £6. Calculate the total variable cost. You are advised to show your workings (2)

14. A furniture store sells chairs for £16 each. In the current month, they sold 350 chairs. Calculate the sales revenue for the month. You are advised to show your workings (2)

15. A coffee shop has recently undertaken some market research. The bar chart below shows the preferred flavour of coffee of each person who responded.



Using the information in the bar chart, identify what percentage of respondents preferred chocolate. You are advised to show your workings (2)

16. The table below contains information about a construction company. Calculate the gross profit margin for the company. You are advised to show your workings (2)

Business	Sales Revenue	Cost of Sales	Gross Profit
Construction	£1,200,000	£720,000	£480,000

17. A service provider completes 123 projects. The fixed costs are £8,000, and the variable cost per project is £50. The business generates £10,455 in sales revenue. Calculate the selling price per unit. You are advised to show your workings (2)

18. A gardening service completes 80 projects. The fixed costs are £4,000, and the variable cost per project is £100. Calculate the total costs. You are advised to show your workings (2)

19. A coffee shop purchases coffee beans from different suppliers. The costs of the coffee beans bought are as follows: £40, £35, £50, £45, and £42. Calculate the average cost of the coffee beans. You are advised to show your workings (2)

20. A retail store received 2,500 customer reviews, out of which 1,800 were positive, 500 were negative, and the remaining were neutral. What percentage of the reviews were positive? You are advised to show your workings (2)

Summary quiz 5 score: /45
